Vedic Maths

Practice Workbook

Write your working in the ruled space under each question.
Answer keys follow each section — attempt first, then check.

$97 \times 96 = 93|12$

$35^2 = 12|25$

$1000 - 357 = 643$

$43 \times 11 = 473$

Original practice material. Exam names are used only to describe the syllabus; no question is an official previous-year question.

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Part XII

Exercise Bank

Over 400 original questions, arranged by topic and difficulty, followed by mixed drills, timed tests and challenge questions. Complete answers with worked solutions are at the end of this part.

Chapter 51: Topic-wise Exercises, Drills and Tests

How to use this bank

- **Topic sections A–H** each have Beginner, Intermediate and Advanced blocks. Finish the Beginner block before moving on.
- **Mixed drills (I)** combine topics, which is how real exams feel.
- **Timed tests (J):** 20 questions in 8 minutes. Write only the answer. Record your score and time in the tracker.
- **Challenge questions (K)** stretch the strongest methods in the book.
- Every answer has a worked solution in the answer key. Read the solution even when you were right; you may find a faster route.

Marking yourself

Tick correct answers, circle mistakes, and write the **reason** for each mistake (concept, careless, slow). Repeat only the circled questions a week later.

A. Numbers and foundations

✎ Numbers and foundations — Beginner

1. In 6,13,08,886, what is the place value of the digit in the hundreds place?

2. Rewrite 61,99,85,668 using international commas (millions).

3. Is 223 prime or composite?

4. Is 32,751 divisible by 9? Use the divisibility rule.

5. Calculate 9 − 12 + 10.

6. How many factors does 48 have? List them.

7. In 7,82,13,000, what is the place value of the digit in the units place?

8. Rewrite 34,88,45,403 using international commas (millions).

Answers: see the answer key (questions 1–8).

✎ Numbers and foundations — Intermediate

9. In 2,97,92,361, what is the place value of the digit in the hundreds place?

10. Rewrite 45,37,04,057 using international commas (millions).

11. Is 299 prime or composite?

12. Is 40,878 divisible by 9? Use the divisibility rule.

13. Calculate (−25) − 13 + (−17).

14. How many factors does 96 have? List them.

15. In 7,16,32,298, what is the place value of the digit in the hundreds place?

16. Rewrite 21,08,38,597 using international commas (millions).

Answers: see the answer key (questions 9–16).

✎ Numbers and foundations — Advanced

17. In 7,54,38,857, what is the place value of the digit in the crores place?

18. Rewrite 30,79,95,041 using international commas (millions).

19. Is 289 prime or composite?

20. Is 4,718 divisible by 4? Use the divisibility rule.

Answers: see the answer key (questions 17–20).

B. Mental addition and subtraction

✎ Mental addition and subtraction — Beginner

21. Add mentally from left to right: 917 + 580

22. 6,616 + 299

23. 1,000 − 605

24. 1,276 − 187

25. Add mentally from left to right: 847 + 467

26. 1,840 + 98

27. 1,00,000 − 73,739

28. 4,691 − 699

29. Add mentally from left to right: 995 + 645

30. 3,113 + 29

Answers: see the answer key (questions 21–30).

✎ Mental addition and subtraction — Intermediate

31. Add: 7,265 + 9,338 + 9,221 + 7,175

32. Estimate 47 × 534 by rounding (approximate answer). *(approximate)*

33. Add: 3,602 + 809 + 8,229 + 9,459

34. Estimate 30 × 400 by rounding (approximate answer). *(approximate)*

35. Add: 6,154 + 297 + 6,795 + 7,966

36. Estimate 88 × 459 by rounding (approximate answer). *(approximate)*

37. Add: 6,743 + 1,341 + 2,079 + 128 + 3,540

38. Estimate 42 × 492 by rounding (approximate answer). *(approximate)*

Answers: see the answer key (questions 31–38).

✎ Mental addition and subtraction — Advanced

39. Add: 2,148 + 4,065 + 547 + 2,122 + 1,142

40. Estimate 82 × 823 by rounding (approximate answer). *(approximate)*

41. Add: 5,557 + 5,404 + 2,841 + 3,285 + 1,081 + 5,930

42. Estimate 54 × 661 by rounding (approximate answer). *(approximate)*

Answers: see the answer key (questions 39–42).

C. Fractions, decimals and percentages

✎ Fractions, decimals and percentages — Beginner

43. Simplify 121/77.

44. Write 3/4 as a percentage.

45. Find 40% of 2,400.

46. Find 11 1/9% of 639 using a fraction.

47. Simplify 28/38.

48. Write 2/5 as a percentage.

49. Find 15% of 3,600.

50. Find 66 2/3% of 216 using a fraction.

51. Simplify 90/171.

52. Write 1/12 as a percentage.

Answers: see the answer key (questions 43–52).

✎ Fractions, decimals and percentages — Intermediate

53. 7/4 + 1

54. A price changes from ₹2,000 to ₹2,500. Find the percentage increase.

55. 5.8 × 7.7

56. 5/7 + 4/11

57. A price changes from ₹400 to ₹440. Find the percentage increase.

58. 4.7 × 0.18

59. 2/5 – 1/6

60. A price changes from ₹2,500 to ₹2,300. Find the percentage decrease.

61. 7.8 × 9.2

62. 1/5 – 3/7

Answers: see the answer key (questions 53–62).

✎ Fractions, decimals and percentages — Advanced

63. 9/5 – 3

64. A price changes from ₹250 to ₹280. Find the percentage increase.

65. 9.7 × 9.8

66. 1 – 5/3

Answers: see the answer key (questions 63–66).

D. Powers, roots and number properties

✎ Powers, roots and number properties — Beginner

67. 47^2

68. 55^2

69. 42^2

70. 24^2

71. 20^2

72. 21^2

73. 56^2

74. 25^2

75. 27^2

76. 33^2

Answers: see the answer key (questions 67–76).

✎ Powers, roots and number properties — Intermediate

77. 14^3

78. $\sqrt{7{,}569}$

79. $\sqrt[3]{27{,}000}$

80. Find the HCF and LCM of 156 and 78.

81. Write 84 as a product of primes.

82. 5^{-4} as a fraction

83. Find the remainder when 34,289 is divided by 7.

84. 19^3

85. $\sqrt{484}$

86. Find the HCF and LCM of 32 and 44.

87. 10^{-2} as a fraction

88. Find the remainder when 88,378 is divided by 9.

Answers: see the answer key (questions 77–88).

✎ Powers, roots and number properties — Advanced

89. Estimate √342 to 2 decimal places (approximate). *(approximate)*

90. Estimate √23 to 2 decimal places (approximate). *(approximate)*

91. Estimate √198 to 2 decimal places (approximate). *(approximate)*

92. Estimate √277 to 2 decimal places (approximate). *(approximate)*

93. Estimate √392 to 2 decimal places (approximate). *(approximate)*

94. Estimate √66 to 2 decimal places (approximate). *(approximate)*

Answers: see the answer key (questions 89–94).

E. Multiplication

✎ Multiplication — Beginner

95. 9,260 × 11

96. 811 × 5

97. 490 × 999

98. 35 × 6

99. 135^2

100. 9,839 × 11

101. 648 × 5

102. 79 × 99

103. 35 × 3

104. 65^2

105. 6,834 × 11

106. 77 × 25

107. 40 × 99

108. 83 × 4

109. 165^2

110. 7,502 × 11

Answers: see the answer key (questions 95–110).

✎ Multiplication — Intermediate

111. 325 × 1,001

112. 995 × 993 (Nikhilam, base 1,000)

113. 112 × 107 (base 100)

114. 37 × 15 (vertically and crosswise)

115. 96 × 94

116. (–27) × 26

117. 963^2

118. 49^2

119. 859 × 1,001

120. 88 × 90 (Nikhilam, base 100)

121. 105 × 101 (base 100)

122. 88 × 31 (vertically and crosswise)

123. 58 × 52

124. 17 × 23

125. 994^2

126. 59^2

127. 51 × 101

128. 998 × 974 (Nikhilam, base 1,000)

129. 112 × 104 (base 100)

130. 62 × 83 (vertically and crosswise)

131. (−19) × 26

132. $1{,}027^2$

Answers: see the answer key (questions 111–132).

✎ Multiplication — Advanced

133. 108 × 97

134. 496 × 508 (working base 500)

135. 314 × 515

136. 495^2 (duplex method)

137. 112 × 98

138. 204 × 204 (working base 200)

139. 420 × 663

140. 281^2 (duplex method)

141. 111 × 91

142. 507 × 502 (working base 500)

143. 714 × 577

144. 502^2 (duplex method)

145. 105 × 96

146. 406 × 399 (working base 400)

Answers: see the answer key (questions 133–146).

F. Division and remainders

✎ Division and remainders — Beginner

147. 1,639 ÷ 23

148. 42,135 ÷ 9 (quotient and remainder)

149. 5,164 ÷ 125 (as a decimal)

150. 7,988 ÷ 20

151. 46,550 ÷ 9 (quotient and remainder)

152. 6,386 ÷ 25 (as a decimal)

153. 5,854 ÷ 9

154. 36,043 ÷ 9 (quotient and remainder)

155. 6,105 ÷ 25 (as a decimal)

156. 2,096 ÷ 9

Answers: see the answer key (questions 147–156).

✎ Division and remainders — Intermediate

157. 3,989 ÷ 13

158. 81,736 ÷ 9 (quotient and remainder)

159. 7,292 ÷ 25 (as a decimal)

160. 25,294 ÷ 997

161. 38,943 ÷ 1012

162. Find the recurring block of 1/29.

Answers: see the answer key (questions 157–162).

✎ Division and remainders — Advanced

163. 42,982 ÷ 996

164. 7,844 ÷ 121

165. Find the recurring block of 1/49.

166. 2,138 ÷ 98

167. 284 ÷ 12

168. Find the recurring block of 1/59.

169. 640 ÷ 8

170. 666 ÷ 14

171. 75 ÷ 7

172. 5,324 ÷ 111

Answers: see the answer key (questions 163–172).

G. Algebra

✎ Algebra — Beginner

173. Solve 6x – 12 = –60

174. Expand (4x – 9)(x – 4)

175. Solve 6x – 20 = –56

176. Expand (4x – 3)(2x – 1)

177. Solve $7x + 3 = -67$

178. Expand $(3x + 7)(x + 3)$

179. Solve $9x + 6 = -30$

180. Expand $(3x - 1)(x - 5)$

Answers: see the answer key (questions 173–180).

✎ Algebra — Intermediate

181. Solve $6x - 5 = 1x - 20$

182. Solve: $9x + 4y = 97$, $2x + 2y = 26$

183. Factorise $x^2 - 8x + 15$

184. $65^2 - 11^2$

185. Use $(a \pm b)^2$ to find 103^2

186. Solve $6x + 2 = 4x + 8$

187. Solve: $9x + 6y = 30$, $1x + 4y = 30$

188. Factorise $x^2 + 15x + 54$

189. $61^2 - 29^2$

190. Use $(a \pm b)^2$ to find 88^2

191. Solve $7x + 7 = 1x - 47$

192. Solve: $9x + 5y = -84$, $6x + 2y = -48$

193. Factorise $x^2 - 16x + 64$

194. $86^2 - 21^2$

195. Use $(a \pm b)^2$ to find 42^2

196. Solve $7x + 11 = 1x + 17$

Answers: see the answer key (questions 181–196).

✎ Algebra — Advanced

197. Solve: $39x + 59y = 214$, $59x + 39y = 374$

198. Solve $(x + 8)(x + 8) = (x + 7)(x + 1)$

199. Solve $x^3 + 3x^2 + 2x + 0 = 0$ by completing the cube

200. Solve $x^2 + 0x - 36 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

201. Solve: $49x + 23y = 455$, $23x + 49y = 481$

202. Solve $(x + 7)(x + 11) = (x + 11)(x + 1)$

203. Solve $x^3 + 12x^2 + 47x + 60 = 0$ by completing the cube

204. Solve $x^2 - 5x - 14 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

205. Solve: $18x + 35y = -176$, $35x + 18y = -142$

206. Solve $(x + 10)(x + 8) = (x + 11)(x + 9)$

207. Solve $x^3 + 6x^2 + 11x + 6 = 0$ by completing the cube

208. Solve $x^2 - 3x + 2 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

Answers: see the answer key (questions 197–208).

H. Arithmetic aptitude

✎ Arithmetic aptitude — Beginner

209. Divide ₹2,061 between A and B in the ratio 5 : 4.

210. Find the average of 59, 54, 50, 47, 54, 60.

211. Find the simple interest on ₹5,000 at 5% per annum for 5 years.

212. A car travels at 72 km/h for 3 hours. Find the distance, and the speed in m/s.

213. Divide ₹6,222 between A and B in the ratio 8 : 9.

214. Find the average of 53, 38, 46, 46, 50, 45, 37.

215. Find the simple interest on ₹15,000 at 12% per annum for 6 years.

216. A car travels at 90 km/h for 2 1/2 hours. Find the distance, and the speed in m/s.

217. Divide ₹3,472 between A and B in the ratio 3 : 5.

218. Find the average of 74, 62, 73, 74, 76, 67.

219. Find the simple interest on ₹40,000 at 12% per annum for 6 years.

220. A car travels at 45 km/h for 3 hours. Find the distance, and the speed in m/s.

Answers: see the answer key (questions 209–220).

✎ Arithmetic aptitude — Intermediate

221. Section A has 28 students with average 59; section B has 16 students with average 64. Find the combined average.

222. An item costing ₹250 is sold for ₹280. Find the profit percentage.

223. Find the single discount equal to successive discounts of 10% and 5%.

224. Find the compound interest on ₹50,000 at 20% per annum, compounded yearly, for 2 years.

225. A can finish a job in 20 days and B in 30 days. How long will they take together?

226. Pipe A fills a tank in 4 hours; pipe B empties it in 6 hours. If both are open, how long to fill the empty tank?

227. Two trains 200 m and 130 m long run at 90 km/h and 45 km/h in the same direction. How long do they take to cross each other?

228. A boat goes downstream at 11 km/h and upstream at 9 km/h. Find the speed of the boat in still water and of the stream.

229. A invests ₹20,000 for 12 months and B invests ₹45,000 for 9 months. Profit is ₹67,596. Find A's share.

230. A father is 3 times as old as his son. In 9 years the sum of their ages will be 62. Find their present ages.

231. Find the last digit of 38^{592}.

232. In how many ways can a committee of 3 be chosen from 9 people?

233. Two dice are thrown. What is the probability that the sum is 10?

234. Volume of a cylinder with radius 28 cm and height 13 cm ($\pi = 22/7$)?

235. Section A has 16 students with average 49; section B has 20 students with average 78. Find the combined average.

236. An item costing ₹600 is sold for ₹528. Find the loss percentage.

237. Find the single discount equal to successive discounts of 30% and 20%.

238. Find the compound interest on ₹20,000 at 20% per annum, compounded yearly, for 2 years.

239. A can finish a job in 15 days and B in 10 days. How long will they take together?

240. Two trains 100 m and 80 m long run at 72 km/h and 36 km/h in opposite directions. How long do they take to cross each other?

241. A invests ₹40,000 for 12 months and B invests ₹35,000 for 10 months. Profit is ₹51,045. Find A's share.

242. A father is 4 times as old as his son. In 2 years the sum of their ages will be 74. Find their present ages.

243. Find the last digit of 38^{239}.

244. In how many ways can a committee of 4 be chosen from 8 people?

245. Two dice are thrown. What is the probability that the sum is 11?

246. Find the area and circumference of a circle of radius 21 cm ($\pi = 22/7$).

247. Section A has 21 students with average 53; section B has 14 students with average 66. Find the combined average.

248. An item costing ₹2,400 is sold for ₹3,360. Find the profit percentage.

249. Find the single discount equal to successive discounts of 20% and 15%.

250. Find the compound interest on ₹40,000 at 10% per annum, compounded yearly, for 2 years.

251. A can finish a job in 6 days and B in 12 days. How long will they take together?

252. Two trains 100 m and 80 m long run at 90 km/h and 45 km/h in the same direction. How long do they take to cross each other?

253. A boat goes downstream at 14 km/h and upstream at 2 km/h. Find the speed of the boat in still water and of the stream.

254. A invests ₹30,000 for 12 months and B invests ₹60,000 for 8 months. Profit is ₹12,551. Find A's share.

Answers: see the answer key (questions 221–254).

✎ Arithmetic aptitude — Advanced

255. In what ratio should rice at ₹20/kg be mixed with rice at ₹70/kg to get a mixture worth ₹33/kg?

256. What day of the week was/is 6 April 2030?

257. In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹54/kg?

258. What day of the week was/is 17 July 1961?

259. In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹48/kg?

260. What day of the week was/is 11 April 1967?

261. In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹64/kg?

262. What day of the week was/is 17 August 1997?

263. In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹61/kg?

264. What day of the week was/is 16 July 1997?

Answers: see the answer key (questions 255–264).

1. Mixed-topic drills

✎ Mixed drill 1

265. In 8,19,85,467, what is the place value of the digit in the tens place?

266. Calculate 19 − 24 + (−28).

267. 1,00,000 − 42,137

268. Simplify 20/85.

269. A price changes from ₹2,000 to ₹2,200. Find the percentage increase.

270. 21^3

271. Write 2,310 as a product of primes.

272. 9,351 × 11

273. 824 × 1,001

274. 144 × 146

275. 397 × 393 (working base 400)

276. 135^2 (duplex method)

Answers: see the answer key (questions 265–276).

✎ Mixed drill 2

277. Rewrite 18,68,02,283 using international commas (millions).

278. 3,403 − 3,127

279. Write 7/8 as a percentage.

280. 4.7 × 2.3

281. √1,369

282. $5^6 \div 5^4$

283. 362 × 5

284. 95 × 98 (Nikhilam, base 100)

285. 145^2

286. 931 × 855

287. 9,481 ÷ 22

288. 766 ÷ 11

Answers: see the answer key (questions 277–288).

✎ Mixed drill 3

289. Is 331 prime or composite?

290. Add mentally from left to right: 355 + 103

291. Add: 9,886 + 7,870 + 4,371 + 9,687

292. Find 5% of 80.

293. Find 16 2/3% of 312 using a fraction.

294. $\sqrt[3]{24{,}389}$

295. Find the remainder when 65,681 is divided by 13.

296. 13 × 99

297. 110 × 104 (base 100)

298. 14 × 16

299. 108^2

300. 71,735 ÷ 9 (quotient and remainder)

Answers: see the answer key (questions 289–300).

✎ Mixed drill 4

301. Is 89,312 divisible by 8? Use the divisibility rule.

302. 2,941 + 197

303. Estimate 72 × 744 by rounding (approximate answer). *(approximate)*

304. 8/11 + 7/12

305. 40^2

306. Find the HCF and LCM of 15 and 24.

307. Estimate √148 to 2 decimal places (approximate). *(approximate)*

308. 16 × 8

309. 26 × 80 (vertically and crosswise)

310. 108 × 93

311. 54^2

312. 3,673 ÷ 50 (as a decimal)

Answers: see the answer key (questions 301–312).

J. Timed mental-math tests (target: 20 questions in 8 minutes; write only the answer)

✎ Timed test I

313. 6,469 × 11

314. 100 − 28

315. 2,747 + 19

316. 588 × 5

317. 44 × 46

318. 89 × 97 (Nikhilam, base 100)

319. Find 30% of 40.

320. 11 × 12

321. 1,194 ÷ 50 (as a decimal)

322. 57^2

323. 10×99

324. Find 37 1/2% of 264 using a fraction.

325. Add mentally from left to right: 858 + 534

326. $40^2 - 16^2$

327. 4,559 × 11

328. 1,00,000 − 93,745

329. 9,309 + 29

330. 929 × 25

331. 132 × 138

332. 95 × 88 (Nikhilam, base 100)

Answers: see the answer key (questions 313–332).

✎ Timed test 2

333. 1,000 − 614

334. 1,347 + 9

335. 139 × 25

336. 970 × 982 (Nikhilam, base 1,000)

337. Find 45% of 120.

338. 2 × 12

339. 6,331 ÷ 5 (as a decimal)

340. 15^2

341. 92 × 99

342. Find 16 2/3% of 648 using a fraction.

343. Add mentally from left to right: 179 + 756

344. $41^2 - 9^2$

345. 4,580 × 11

346. 1,00,000 − 21,843

347. 8,548 + 9

348. 816 × 125

349. 32 × 38

350. 88 × 99 (Nikhilam, base 100)

351. Find 75% of 160.

352. 12 × 4

Answers: see the answer key (questions 333–352).

✎ Timed test 3

353. 1,000 − 151

354. 7,746 + 9

355. 541 × 5

356. 49 × 41

357. 92 × 99 (Nikhilam, base 100)

358. Find 30% of 1,200.

359. 9 × 6

360. 2,530 ÷ 50 (as a decimal)

361. 22^2

362. 391 × 999

363. Find 37 1/2% of 232 using a fraction.

364. Add mentally from left to right: 741 + 547

365. $92^2 - 29^2$

366. 7,707 × 11

367. 115^2

368. 100 − 27

369. 2,231 + 199

370. 717 × 125

371. 81 × 89

372. 96 × 98 (Nikhilam, base 100)

Answers: see the answer key (questions 353–372).

✎ Timed test 4

373. 5,005 + 99

374. 216 × 5

375. 119 × 111

376. 983 × 969 (Nikhilam, base 1,000)

377. Find 20% of 4,800.

378. 10 × 12

379. 6,644 ÷ 5 (as a decimal)

380. 29^2

381. 57 × 99

382. Find 16 2/3% of 768 using a fraction.

383. Add mentally from left to right: 489 + 105

384. $46^2 - 20^2$

385. 8,512 × 11

386. 10,000 − 9,585

387. 1,593 + 29

388. 763 × 50

389. 108 × 102

390. 97 × 94 (Nikhilam, base 100)

391. Find 20% of 40.

392. 9 × 4

Answers: see the answer key (questions 373–392).

K. Challenge questions

✎ Challenge set

393. Estimate √336 to 2 decimal places (approximate). *(approximate)*

394. 110 × 90

395. 408 × 398 (working base 400)

396. 145 × 403

397. 851^2 (duplex method)

398. 679 ÷ 8

399. 926 ÷ 13

400. Find the recurring block of 1/39.

401. Solve: 22x + 32y = 366, 32x + 22y = 336

402. Solve $(x + 3)(x + 11) = (x + 12)(x + 1)$

403. Solve $x^3 + 9x^2 + 26x + 24 = 0$ by completing the cube

404. Solve $x^2 + 4x + 0 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

405. In what ratio should rice at ₹30/kg be mixed with rice at ₹70/kg to get a mixture worth ₹32/kg?

406. What day of the week was/is 10 March 1976?

407. Estimate $\sqrt{281}$ to 2 decimal places (approximate). *(approximate)*

408. 103×94

409. 308×303 (working base 300)

410. 970×783

411. 200^2 (duplex method)

412. $459 \div 7$

Answers: see the answer key (questions 393–412).

Part XII answer key with worked solutions

1. **800** — The digit is 8; hundreds = 100; 8 × 100 = 800.
2. **619,985,668** — International grouping uses threes from the right: thousand, million, billion.
3. **Prime** — √223 ≈ 14. No prime up to 14 divides 223, so it is prime.
4. **Yes** — Rule for 9: digit sum divisible by 9. 32,751 ÷ 9 leaves remainder 0.
5. **7** — Subtracting 12 is the same as adding −12. 9 + −12 + 10 = 7.
6. **10: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48** — Pair factors from the outside in: 1×48, 2×24, 3×16, 4×12, 6×8.
7. **0** — The digit is 0; units = 1; 0 × 1 = 0.
8. **348,845,403** — International grouping uses threes from the right: thousand, million, billion.
9. **300** — The digit is 3; hundreds = 100; 3 × 100 = 300.
10. **453,704,057** — International grouping uses threes from the right: thousand, million, billion.
11. **Composite** — 299 = 13 × 23, so it is composite.
12. **Yes** — Rule for 9: digit sum divisible by 9. 40,878 ÷ 9 leaves remainder 0.
13. **−55** — Subtracting 13 is the same as adding −13. −25 + −13 + −17 = −55.
14. **12: 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96** — Pair factors from the outside in: 1×96, 2×48, 3×32, 4×24, 6×16, 8×12.
15. **200** — The digit is 2; hundreds = 100; 2 × 100 = 200.
16. **210,838,597** — International grouping uses threes from the right: thousand, million, billion.
17. **7,00,00,000** — The digit is 7; crores = 1,00,00,000; 7 × 1,00,00,000 = 7,00,00,000.
18. **307,995,041** — International grouping uses threes from the right: thousand, million, billion.
19. **Composite** — 289 = 17 × 17, so it is composite.
20. **No** — Rule for 4: last two digits divisible by 4. 4,718 ÷ 4 leaves remainder 2.
21. **1,497** — Hundreds 1400, tens 90, units 7: 1400 + 90 + 7 = 1497.
22. **6,915** — 299 = 300 − 1. 6,616 + 300 = 6,916, then − 1 = 6,915.
23. **395** — All from 9, last from 10: 9−6=3 9−0=9, 10−5=5 → 395.
24. **1,089** — Subtract 190 instead: 1,276 − 190 = 1,086; add back 3 → 1,089.
25. **1,314** — Hundreds 1200, tens 100, units 14: 1200 + 100 + 14 = 1314.
26. **1,938** — 98 = 100 − 2. 1,840 + 100 = 1,940, then − 2 = 1,938.
27. **26,261** — All from 9, last from 10: 9−7=2 9−3=6 9−7=2 9−3=6, 10−9=1 → 26261.
28. **3,992** — Subtract 700 instead: 4,691 − 700 = 3,991; add back 1 → 3,992.
29. **1,640** — Hundreds 1500, tens 130, units 10: 1500 + 130 + 10 = 1640.
30. **3,142** — 29 = 30 − 1. 3,113 + 30 = 3,143, then − 1 = 3,142.
31. **32,999** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 5.
32. **≈ 25,000 (exact 25,098)** — 47 ≈ 50, 534 ≈ 500; 50 × 500 = 25,000. This is an estimate, not the exact product.
33. **22,099** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 4.
34. **≈ 12,000 (exact 12,000)** — 30 ≈ 30, 400 ≈ 400; 30 × 400 = 12,000. This is an estimate, not the exact product.
35. **21,212** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 8.
36. **≈ 41,400 (exact 40,392)** — 88 ≈ 90, 459 ≈ 460; 90 × 460 = 41,400. This is an estimate, not the exact product.
37. **13,831** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 7.
38. **≈ 19,600 (exact 20,664)** — 42 ≈ 40, 492 ≈ 490; 40 × 490 = 19,600. This is an estimate, not the exact product.
39. **10,024** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 7.
40. **≈ 64,000 (exact 67,486)** — 82 ≈ 80, 823 ≈ 800; 80 × 800 = 64,000. This is an estimate, not the exact product.
41. **24,098** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 5.
42. **≈ 35,000 (exact 35,694)** — 54 ≈ 50, 661 ≈ 700; 50 × 700 = 35,000. This is an estimate, not the exact product.
43. **11/7** — HCF of 121 and 77 is 11. Divide both by 11.
44. **75%** — 3/4 × 100 = 75%.
45. **960** — Split 40% into easy parts (10%, 5%, 1% …) or use fractions: 40% = 2/5. Result 960.

46. **71** — 11 1/9% = 1/9. 1/9 × 639 = 71.

47. **14/19** — HCF of 28 and 38 is 2. Divide both by 2.

48. **40%** — 2/5 × 100 = 40%.

49. **540** — Split 15% into easy parts (10%, 5%, 1% …) or use fractions: 15% = 3/20. Result 540.

50. **144** — 66 2/3% = 2/3. 2/3 × 216 = 144.

51. **10/19** — HCF of 90 and 171 is 9. Divide both by 9.

52. **8 1/3%** — 1/12 × 100 = 8 1/3%.

53. **2 3/4** — LCM of denominators = 4. Convert, then add numerators and simplify: 11/4.

54. **25%** — Change = 500. % change = change ÷ original × 100 = 500/2000 × 100 = 25%.

55. **44.66** — Multiply without points: 58 × 77 = 4466. Decimal places: 1 + 1 = 2, so the answer is 44.66.

56. **1 6/77** — LCM of denominators = 77. Convert, then add numerators and simplify: 83/77.

57. **10%** — Change = 40. % change = change ÷ original × 100 = 40/400 × 100 = 10%.

58. **0.846** — Multiply without points: 47 × 18 = 846. Decimal places: 1 + 2 = 3, so the answer is 0.846.

59. **7/30** — LCM of denominators = 30. Convert, then subtract numerators and simplify: 7/30.

60. **8%** — Change = 200. % change = change ÷ original × 100 = 200/2500 × 100 = 8%.

61. **71.76** — Multiply without points: 78 × 92 = 7176. Decimal places: 1 + 1 = 2, so the answer is 71.76.

62. **–8/35** — LCM of denominators = 35. Convert, then subtract numerators and simplify: –8/35.

63. **–1 1/5** — LCM of denominators = 5. Convert, then subtract numerators and simplify: –6/5.

64. **12%** — Change = 30. % change = change ÷ original × 100 = 30/250 × 100 = 12%.

65. **95.06** — Multiply without points: 97 × 98 = 9506. Decimal places: 1 + 1 = 2, so the answer is 95.06.

66. **–2/3** — LCM of denominators = 3. Convert, then subtract numerators and simplify: –2/3.

67. **2,209** — 47 × 47 = 2209.

68. **3,025** — 55 × 55 = 3025.

69. **1,764** — 42 × 42 = 1764.

70. **576** — 24 × 24 = 576.

71. **400** — 20 × 20 = 400.

72. **441** — 21 × 21 = 441.

73. **3,136** — 56 × 56 = 3136.

74. **625** — 25 × 25 = 625.

75. **729** — 27 × 27 = 729.

76. **1,089** — 33 × 33 = 1089.

77. **2,744** — 14 × 14 × 14 = 2,744.

78. **87** — Last digit 9 → root ends in 3 or 7. Leading part 75 lies between 8^2 and 9^2, so tens digit is 8. Check 85^2 = 7225 to choose → 87.

79. **30** — Last digit 0 → cube root ends in 0 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 27 lies between 3^3 and 4^3 → tens digit 3.

80. **HCF 78, LCM 156** — HCF × LCM = product: 78 × 156 = 12168 = 156 × 78.

81. **$2^2 \times 3 \times 7$** — Divide repeatedly by the smallest prime that works (factor tree).

82. **1/625** — A negative exponent means reciprocal: $1/5^4$.

83. **3** — 34,289 = 7 × 4,898 + 3.

84. **6,859** — 19 × 19 × 19 = 6,859.

85. **22** — Last digit 4 → root ends in 2 or 8. Leading part 4 lies between 2^2 and 3^2, so tens digit is 2. Check 25^2 = 625 to choose → 22.

86. **HCF 4, LCM 352** — HCF × LCM = product: 4 × 352 = 1408 = 32 × 44.

87. **1/100** — A negative exponent means reciprocal: $1/10^2$.

88. **7** — 88,378 = 9 × 9,819 + 7. Shortcut: digit sum mod 9.

89. **≈ 18.50 (true value 18.493)** — Nearest perfect square 324 = 18^2. $\sqrt{n} \approx 18 + (342 - 324) \div (2 \times 18) = 18.500$.

90. **≈ 4.80 (true value 4.796)** — Nearest perfect square 25 = 5^2. $\sqrt{n} \approx 5 + (23 - 25) \div (2 \times 5) = 4.800$.

91. **≈ 14.07 (true value 14.071)** — Nearest perfect square 196 = 14^2. $\sqrt{n} \approx 14 + (198 - 196) \div (2 \times 14) = 14.071$.

92. **≈ 16.65 (true value 16.643)** — Nearest perfect square 289 = 17^2. $\sqrt{n} \approx 17 + (277 - 289) \div (2 \times 17) = 16.647$.

93. **≈ 19.80 (true value 19.799)** — Nearest perfect square 400 = 20^2. $\sqrt{n} \approx 20 + (392 - 400) \div (2 \times 20) = 19.800$.

94. **≈ 8.12 (true value 8.124)** — Nearest perfect square 64 = 8^2. $\sqrt{n} \approx 8 + (66 - 64) \div (2 \times 8) = 8.125$.

95. **1,01,860** — Write first digit 9 and last digit 0 In between, add neighbours: 9+2=11, 2+6=8, 6+0=6 Carry where a sum ≥ 10 Answer = 1,01,860

96. **4,055** — 5 = 10 / 2 → 811 × 10 ÷ 2 = 4,055.

97. **4,89,510** — Left: 490 − 1 = 489 Right: 999 − 489 = 510 Answer = 4,89,510

98. **210** — (30 + 5) × 6 = 180 + 30 = 210.

99. **18,225** — Digits before 5: 13 13 × (one more = 14) = 182 Attach 25 → 18,225

100. **1,08,229** — Write first digit 9 and last digit 9 In between, add neighbours: 9+8=17, 8+3=11, 3+9=12 Carry where a sum ≥ 10 Answer = 1,08,229

101. **3,240** — 5 = 10 / 2 → 648 × 10 ÷ 2 = 3,240.

102. **7,821** — Left: 79 − 1 = 78 Right: 99 − 78 = 21 Answer = 7,821

103. **105** — (30 + 5) × 3 = 90 + 15 = 105.

104. **4,225** — Digits before 5: 6 6 × (one more = 7) = 42 Attach 25 → 4,225

105. **75,174** — Write first digit 6 and last digit 4 In between, add neighbours: 6+8=14, 8+3=11, 3+4=7 Carry where a sum ≥ 10 Answer = 75,174

106. **1,925** — 25 = 100 / 4 → 77 × 100 ÷ 4 = 1,925.

107. **3,960** — Left: 40 − 1 = 39 Right: 99 − 39 = 60 Answer = 3,960

108. **332** — (80 + 3) × 4 = 320 + 12 = 332.

109. **27,225** — Digits before 5: 16 16 × (one more = 17) = 272 Attach 25 → 27,225

110. **82,522** — Write first digit 7 and last digit 2 In between, add neighbours: 7+5=12, 5+0=5, 0+2=2 Carry where a sum ≥ 10 Answer = 82,522

111. **3,25,325** — 1001 = 1000 + 1, so write 325 twice (padded): 325 × 1001 = 3,25,325.

112. **9,88,035** — Base = 1,000 Deviations: 995 → −5, 993 → −7 Left part (cross): 995 − 7 = 988 Right part: (−5) × (−7) = 35 Right part padded to 3 digit(s): 035 Answer = 9,88,035

113. **11,984** — Base = 100 Deviations: 112 → +12, 107 → +7 Left part (cross): 112 + 7 = 119 Right part: (+12) × (+7) = 84 Right part padded to 2 digit(s): 84 Answer = 11,984

114. **555** — Column 1 (from right): 7×5 = 35 → write 5, carry 3 Column 2 (from right): 7×1 + 3×5 = 22, + carry 3 = 25 → write 5, carry 2 Column 3 (from right): 3×1 = 3, + carry 2 = 5 → write 5, carry 0 Answer = 555

115. **9,024** — Same leading part 9; last digits 6 + 4 = 10 ✓ Left: 9 × 10 = 90 Right: 6 × 4 = 24 (always two digits) Answer = 9,024

116. **−702** — Same signs → positive; different signs → negative. 27 × 26 = 702.

117. **9,27,369** — Base 1,000, deviation −37 Left: 963 − 37 = 926 Right: 37^2 = 1369 (needs 3 digit(s), carry 1) Answer = 9,27,369

118. **2,401** — Deviation from 50: −1 Left: 25 − 1 = 24 Right: 1^2 = 01 Answer = 2,401

119. **8,59,859** — 1001 = 1000 + 1, so write 859 twice (padded): 859 × 1001 = 8,59,859.

120. **7,920** — Base = 100 Deviations: 88 → −12, 90 → −10 Left part (cross): 88 − 10 = 78 Right part: (−12) × (−10) = 120 Right part has more than 2 digit(s): carry 1 to the left Answer = 7,920

121. **10,605** — Base = 100 Deviations: 105 → +5, 101 → +1 Left part (cross): 105 + 1 = 106 Right part: (+5) × (+1) = 5 Right part padded to 2 digit(s): 05 Answer = 10,605

122. **2,728** — Column 1 (from right): 8×1 = 8 → write 8, carry 0 Column 2 (from right): 8×3 + 8×1 = 32 → write 2, carry 3 Column 3 (from right): 8×3 = 24, + carry 3 = 27 → write 7, carry 2 Answer = 2,728

123. **3,016** — Same leading part 5; last digits 8 + 2 = 10 ✓ Left: 5 × 6 = 30 Right: 8 × 2 = 16 (always two digits) Answer = 3,016

124. **391** — Same signs → positive; different signs → negative. 17 × 23 = 391.

125. **9,88,036** — Base 1,000, deviation −6 Left: 994 − 6 = 988 Right: 6^2 = 36 (needs 3 digit(s)) Answer = 9,88,036

126. **3,481** — Deviation from 50: +9 Left: 25 + 9 = 34 Right: 9^2 = 81 Answer = 3,481

127. **5,151** — 101 = 100 + 1, so write 51 twice (padded): 51 × 101 = 5,151.

128. **9,72,052** — Base = 1,000 Deviations: 998 → –2, 974 → –26 Left part (cross): 998 – 26 = 972 Right part: (–2) × (–26) = 52 Right part padded to 3 digit(s): 052 Answer = 9,72,052

129. **11,648** — Base = 100 Deviations: 112 → +12, 104 → +4 Left part (cross): 112 + 4 = 116 Right part: (+12) × (+4) = 48 Right part padded to 2 digit(s): 48 Answer = 11,648

130. **5,146** — Column 1 (from right): 2×3 = 6 → write 6, carry 0 Column 2 (from right): 2×8 + 6×3 = 34 → write 4, carry 3 Column 3 (from right): 6×8 = 48, + carry 3 = 51 → write 1, carry 5 Answer = 5,146

131. **–494** — Same signs → positive; different signs → negative. 19 × 26 = 494.

132. **10,54,729** — Base 1,000, deviation +27 Left: 1027 + 27 = 1054 Right: 27^2 = 729 (needs 3 digit(s)) Answer = 10,54,729

133. **10,476** — Base = 100 Deviations: 108 → +8, 97 → –3 Left part (cross): 108 – 3 = 105 Right part: (+8) × (–3) = –24 Right part is negative: borrow 1 from left → 105 × 100 – 24 Answer = 10,476

134. **2,51,968** — Working base 500 = 5 × 100. Deviations –4, +8. Left: (496 +8) × 5 = 2520. Right: –4 × 8 = –32. 2520 × 100 + –32 = 2,51,968.

135. **1,61,710** — Column 1 (from right): 4×5 = 20 → write 0, carry 2 Column 2 (from right): 4×1 + 1×5 = 9, + carry 2 = 11 → write 1, carry 1 Column 3 (from right): 4×5 + 1×1 + 3×5 = 36, + carry 1 = 37 → write 7, carry 3 Column 4 (from right): 1×5 + 3×1 = 8, + carry 3 = 11 → write 1, carry 1 Column 5 (from right): 3×5 = 15, + carry 1 = 16 → write 6, carry 1 Answer = 1,61,710

136. **2,45,025** — Duplexes from left: 16, 72, 121, 90, 25 Combine with carries (right to left) → 2,45,025

137. **10,976** — Base = 100 Deviations: 112 → +12, 98 → –2 Left part (cross): 112 – 2 = 110 Right part: (+12) × (–2) = –24 Right part is negative: borrow 1 from left → 110 × 100 – 24 Answer = 10,976

138. **41,616** — Working base 200 = 2 × 100. Deviations +4, +4. Left: (204 +4) × 2 = 416. Right: 4 × 4 = 16. 416 × 100 + 16 = 41,616.

139. **2,78,460** — Column 1 (from right): 0×3 = 0 → write 0, carry 0 Column 2 (from right): 0×6 + 2×3 = 6 → write 6, carry 0 Column 3 (from right): 0×6 + 2×6 + 4×3 = 24 → write 4, carry 2 Column 4 (from right): 2×6 + 4×6 = 36, + carry 2 = 38 → write 8, carry 3 Column 5 (from right): 4×6 = 24, + carry 3 = 27 → write 7, carry 2 Answer = 2,78,460

140. **78,961** — Duplexes from left: 4, 32, 68, 16, 1 Combine with carries (right to left) → 78,961

141. **10,101** — Base = 100 Deviations: 111 → +11, 91 → –9 Left part (cross): 111 – 9 = 102 Right part: (+11) × (–9) = –99 Right part is negative: borrow 1 from left → 102 × 100 – 99 Answer = 10,101

142. **2,54,514** — Working base 500 = 5 × 100. Deviations +7, +2. Left: (507 +2) × 5 = 2545. Right: 7 × 2 = 14. 2545 × 100 + 14 = 2,54,514.

143. **4,11,978** — Column 1 (from right): 4×7 = 28 → write 8, carry 2 Column 2 (from right): 4×7 + 1×7 = 35, + carry 2 = 37 → write 7, carry 3 Column 3 (from right): 4×5 + 1×7 + 7×7 = 76, + carry 3 = 79 → write 9, carry 7 Column 4 (from right): 1×5 + 7×7 = 54, + carry 7 = 61 → write 1, carry 6 Column 5 (from right): 7×5 = 35, + carry 6 = 41 → write 1, carry 4 Answer = 4,11,978

144. **2,52,004** — Duplexes from left: 25, 0, 20, 0, 4 Combine with carries (right to left) → 2,52,004

145. **10,080** — Base = 100 Deviations: 105 → +5, 96 → –4 Left part (cross): 105 – 4 = 101 Right part: (+5) × (–4) = –20 Right part is negative: borrow 1 from left → 101 × 100 – 20 Answer = 10,080

146. **1,61,994** — Working base 400 = 4 × 100. Deviations +6, –1. Left: (406 –1) × 4 = 1620. Right: 6 × –1 = –6. 1620 × 100 + –6 = 1,61,994.

147. **Q = 71, R = 6** — Check: 23 × 71 + 6 = 1,639.

148. **Q = 4,681, R = 6** — Running digit sums (quotient columns): 4, 6, 7, 10 Last running sum + last digit = remainder column 15 Adjust carries/remainder ≥ 9 → quotient 4,681, remainder 6

149. **41.312** — ÷ 125 = × 8 ÷ 1000: 5,164 × 8 = 41,312, ÷ 1000 = 41.312.

150. **Q = 399, R = 8** — Check: 20 × 399 + 8 = 7,988.

151. **Q = 5,172, R = 2** — Running digit sums (quotient columns): 4, 10, 15, 20 Last running sum + last digit = remainder column 20 Adjust carries/remainder ≥ 9 → quotient 5,172, remainder 2

152. **255.44** — ÷ 25 = × 4 ÷ 100: 6,386 × 4 = 25,544, ÷ 100 = 255.44.

153. **Q = 650, R = 4** — Check: 9 × 650 + 4 = 5,854.

154. **Q = 4,004, R = 7** — Running digit sums (quotient columns): 3, 9, 9, 13 Last running sum + last digit = remainder column 16 Adjust carries/remainder ≥ 9 → quotient 4,004, remainder 7

155. **244.2** — ÷ 25 = × 4 ÷ 100: 6,105 × 4 = 24,420, ÷ 100 = 244.2.

156. **Q = 232, R = 8** — Check: 9 × 232 + 8 = 2,096.

157. **Q = 306, R = 11** — Check: 13 × 306 + 11 = 3,989.

158. **Q = 9,081, R = 7** — Running digit sums (quotient columns): 8, 9, 16, 19 Last running sum + last digit = remainder column 25 Adjust carries/remainder ≥ 9 → quotient 9,081, remainder 7

159. **291.68** — ÷ 25 = × 4 ÷ 100: 7,292 × 4 = 29,168, ÷ 100 = 291.68.

160. **Q = 25, R = 369** — Divisor 997 = 1,000 − 3; write complement digits 003 Split dividend: 25 | 294 Column totals after multiplying by the complement: [2, 5] | [2, 15, 19] Resolve carries and remainder → quotient 25, remainder 369

161. **Q = 38, R = 487** — Divisor 1012 = 1,000 + 12; transposed flags: −0, −1, −2 Column values after applying flags: [3, 8] | [6, -10, -13] Resolve negatives/carries → quotient 38, remainder 487

162. **0.(034482758620689655172413…)** — Ekadhika (one more than 2) = 3 Start with 1 on the right; repeatedly multiply by 3, write the unit digit, carry the rest Recurring block (28 digits): 034482758620689655172413…

163. **Q = 43, R = 154** — Divisor 996 = 1,000 − 4; write complement digits 004 Split dividend: 42 | 982 Column totals after multiplying by the complement: [4, 2] | [9, 24, 10] Resolve carries and remainder → quotient 43, remainder 154

164. **Q = 64, R = 100** — Divisor 121 = 100 + 21; transposed flags: −2, −1 Column values after applying flags: [7, -6] | [9, 10] Resolve negatives/carries → quotient 64, remainder 100

165. **0.(020408163265306122448979…)** — Ekadhika (one more than 4) = 5 Start with 1 on the right; repeatedly multiply by 5, write the unit digit, carry the rest Recurring block (42 digits): 020408163265306122448979…

166. **Q = 21, R = 80** — Divisor 98 = 100 − 2; write complement digits 02 Split dividend: 21 | 38 Column totals after multiplying by the complement: [2, 1] | [7, 10] Resolve carries and remainder → quotient 21, remainder 80

167. **Q = 23, R = 8** — Divisor 12 = 10 + 2; transposed flags: −2 Column values after applying flags: [2, 4] | [-4] Resolve negatives/carries → quotient 23, remainder 8

168. **0.(016949152542372881355932…)** — Ekadhika (one more than 5) = 6 Start with 1 on the right; repeatedly multiply by 6, write the unit digit, carry the rest Recurring block (58 digits): 016949152542372881355932…

169. **Q = 80, R = 0** — Divisor 8 = 10 − 2; write complement digits 2 Split dividend: 64 | 0 Column totals after multiplying by the complement: [6, 16] | [32] Resolve carries and remainder → quotient 80, remainder 0

170. **Q = 47, R = 8** — Divisor 14 = 10 + 4; transposed flags: −4 Column values after applying flags: [6, -18] | [78] Resolve negatives/carries → quotient 47, remainder 8

171. **Q = 10, R = 5** — Divisor 7 = 10 − 3; write complement digits 3 Split dividend: 7 | 5 Column totals after multiplying by the complement: [7] | [26] Resolve carries and remainder → quotient 10, remainder 5

172. **Q = 47, R = 107** — Divisor 111 = 100 + 11; transposed flags: −1, −1 Column values after applying flags: [5, -2] | [-1, 6] Resolve negatives/carries → quotient 47, remainder 107

173. **x = −8** — Transpose −12 (change sign): 6x = −60 + 12 = −48; divide by 6: x = −8.

174. **$4x^2 - 25x + 36$** — Vertical (left): 4 × 1 = 4. Crosswise: 4 × (−4) + 1 × (−9) = −25. Vertical (right): (−9) × (−4) = 36.

175. **x = −6** — Transpose −20 (change sign): 6x = −56 + 20 = −36; divide by 6: x = −6.

176. **$8x^2 - 10x + 3$** — Vertical (left): 4 × 2 = 8. Crosswise: 4 × (−1) + 2 × (−3) = −10. Vertical (right): (−3) × (−1) = 3.

177. **x = −10** — Transpose 3 (change sign): 7x = −67 − 3 = −70; divide by 7: x = −10.

178. **$3x^2 + 16x + 21$** — Vertical (left): 3 × 1 = 3. Crosswise: 3 × (3) + 1 × (7) = 16. Vertical (right): (7) × (3) = 21.

179. **x = −4** — Transpose 6 (change sign): 9x = −30 − 6 = −36; divide by 9: x = −4.

180. **$3x^2 - 16x + 5$** — Vertical (left): 3 × 1 = 3. Crosswise: 3 × (−5) + 1 × (−1) = −16. Vertical (right): (−1) × (−5) = 5.

181. **x = −3** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−20 − −5)/(6 − 1) = −15/5 = −3.

182. **x = 9, y = 4** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (194 - 104)/10 = 9$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (234 - 194)/10 = 4$.

183. **(x − 5)(x − 3)** — Find two numbers with product 15 and sum −8: −5 and −3. Check with Gunita-samuccaya (put x = 1): (1 − 5)(1 − 3) = 8 and 1 − 8 + 15 = 8.

184. **4,104** — $a^2 - b^2 = (a + b)(a - b) = 76 \times 54 = 4{,}104$.

185. **10,609** — $(100 + 3)^2 = 10000 + 2{\times}100{\times}3 + 9 = 10{,}609$.

186. **x = 3** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (8 − 2)/(6 − 4) = 6/2 = 3.

187. **x = −2, y = 8** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (120 - 180)/30 = -2$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (270 - 30)/30 = 8$.

188. **(x + 6)(x + 9)** — Find two numbers with product 54 and sum 15: 6 and 9. Check with Gunita-samuccaya (put x = 1): (1 + 6)(1 + 9) = 70 and 1 + 15 + 54 = 70.

189. **2,880** — $a^2 - b^2 = (a + b)(a - b) = 90 \times 32 = 2{,}880$.

190. **7,744** — $(90 - 2)^2 = 8100 - 2\times90\times2 + 4 = 7{,}744$.

191. **x = –9** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–47 – 7)/(7 – 1) = –54/6 = –9.

192. **x = –6, y = –6** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-168 - -240)/-12 = -6$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (-432 - -504)/-12 = -6$.

193. **(x – 8)(x – 8)** — Find two numbers with product 64 and sum –16: –8 and –8. Check with Gunita-samuccaya (put x = 1): (1 – 8)(1 – 8) = 49 and 1 – 16 + 64 = 49.

194. **6,955** — $a^2 - b^2 = (a + b)(a - b) = 107 \times 65 = 6{,}955$.

195. **1,764** — $(40 + 2)^2 = 1600 + 2\times40\times2 + 4 = 1{,}764$.

196. **x = 1** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (17 – 11)/(7 – 1) = 6/6 = 1.

197. **x = 7, y = –1** — Add: 98(x + y) = 588 → x + y = 6. Subtract: -20(x – y) = –160 → x – y = 8. Solve: x = 7, y = –1.

198. **x = –7 1/8** — x = (cd – ab)/(a + b – c – d) = (7 – 64)/(16 – 8).

199. **x = –2, –1, 0** — $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$. The given cubic = $(x + 1)^3 - (x + 1)$. Let y = x + 1: $y^3 - y = 0 \rightarrow y = 0, \pm1$.

200. **x = 6 or –6** — Differential: 2x + 0. Discriminant = 0 – -144 = 144, √ = 12. 2x + 0 = ±12.

201. **x = 6, y = 7** — Add: 72(x + y) = 936 → x + y = 13. Subtract: 26(x – y) = –26 → x – y = –1. Solve: x = 6, y = 7.

202. **x = –11** — x = (cd – ab)/(a + b – c – d) = (11 – 77)/(18 – 12).

203. **x = –5, –4, –3** — $(x + 4)^3 = x^3 + 12x^2 + 48x + 64$. The given cubic = $(x + 4)^3 - (x + 4)$. Let y = x + 4: $y^3 - y = 0 \rightarrow y = 0, \pm1$.

204. **x = 7 or –2** — Differential: 2x – 5. Discriminant = 25 – -56 = 81, √ = 9. 2x – 5 = ±9.

205. **x = –2, y = –4** — Add: 53(x + y) = –318 → x + y = –6. Subtract: -17(x – y) = –34 → x – y = 2. Solve: x = –2, y = –4.

206. **x = –9 1/2** — x = (cd – ab)/(a + b – c – d) = (99 – 80)/(18 – 20).

207. **x = –3, –2, –1** — $(x + 2)^3 = x^3 + 6x^2 + 12x + 8$. The given cubic = $(x + 2)^3 - (x + 2)$. Let y = x + 2: $y^3 - y = 0 \rightarrow y = 0, \pm1$.

208. **x = 2 or 1** — Differential: 2x – 3. Discriminant = 9 – 8 = 1, √ = 1. 2x – 3 = ±1.

209. **A ₹1,145, B ₹916** — Total parts 9; one part = 2,061 ÷ 9 = 229.

210. **54** — Assume 55. Deviations: +4, –1, –5, –8, –1, +5; total –6; ÷ 6 = –1. Average = 54.

211. **₹1,250** — SI = PRT/100 = 5,000 × 5 × 5 / 100 = 1,250.

212. **216 km; 20 m/s** — D = S × T = 72 × 3 = 216 km. km/h → m/s: × 5/18 → 20 m/s.

213. **A ₹2,928, B ₹3,294** — Total parts 17; one part = 6,222 ÷ 17 = 366.

214. **45** — Assume 44. Deviations: +9, –6, +2, +2, +6, +1, –7; total +7; ÷ 7 = 1. Average = 45.

215. **₹10,800** — SI = PRT/100 = 15,000 × 12 × 6 / 100 = 10,800.

216. **225 km; 25 m/s** — D = S × T = 90 × 2 1/2 = 225 km. km/h → m/s: × 5/18 → 25 m/s.

217. **A ₹1,302, B ₹2,170** — Total parts 8; one part = 3,472 ÷ 8 = 434.

218. **71** — Assume 70. Deviations: +4, –8, +3, +4, +6, –3; total +6; ÷ 6 = 1. Average = 71.

219. **₹28,800** — SI = PRT/100 = 40,000 × 12 × 6 / 100 = 28,800.

220. **135 km; 12 1/2 m/s** — D = S × T = 45 × 3 = 135 km. km/h → m/s: × 5/18 → 12 1/2 m/s.

221. **60.82** — (28×59 + 16×64) ÷ 44 = 2676 ÷ 44.

222. **12% profit** — Profit = 30. % = 30/250 × 100 = 12%.

223. **14 1/2%** — Net = a + b – ab/100 = 10 + 5 – 50/100 = 14 1/2%.

224. **₹22,000** — $A = P(1 + R/100)^n = 50{,}000 \times (120/100)^2 = ₹72{,}000$. CI = A – P = ₹22,000.

225. **12 days** — Total work = LCM(20,30) = 60 units. A does 3/day, B does 2/day; together 5/day. Time = 60/5 = 12 days.

226. **12 hours** — Net rate = 1/4 – 1/6 = 1/12 tank/hour, so time = 12 hours.

227. **26 2/5 s** — Relative speed = 45 km/h = 12 1/2 m/s. Distance = 200 + 130 = 330 m. Time = 26 2/5 s.

228. **Boat 10 km/h, stream 1 km/h** — Boat = (down + up)/2 = (11 + 9)/2 = 10. Stream = (down – up)/2 = 1.

229. **₹25,152** — Ratio = 20,000×12 : 45,000×9 = 16 : 27. A's share = 16/43 × 67,596 = 25,152.

230. **Son 11, father 33** — Let son = x, father = 3x. (3x + 9) + (x + 9) = 62 → 4x = 44 → x = 11.

231. **6** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 592 mod 4 = 4 → 6.

232. **84** — Order does not matter: ${}^nC_r = 9!/(3!\cdot6!) = 84$.

233. **1/12** — 3 favourable outcomes out of 36 → 3/36 = 1/12.

234. **32,032 cm³** — V = πr²h = 22/7 × 784 × 13 = 32,032.

235. **65.11** — (16×49 + 20×78) ÷ 36 = 2344 ÷ 36.

236. **12% loss** — Loss = 72. % = 72/600 × 100 = 12%.

237. **44%** — Net = a + b − ab/100 = 30 + 20 − 600/100 = 44%.

238. **₹8,800** — $A = P(1 + R/100)^n = 20{,}000 \times (120/100)^2 = ₹28{,}800$. CI = A − P = ₹8,800.

239. **6 days** — Total work = LCM(15,10) = 30 units. A does 2/day, B does 3/day; together 5/day. Time = 30/5 = 6 days.

240. **6 s** — Relative speed = 108 km/h = 30 m/s. Distance = 100 + 80 = 180 m. Time = 6 s.

241. **₹29,520** — Ratio = 40,000×12 : 35,000×10 = 48 : 35. A's share = 48/83 × 51,045 = 29,520.

242. **Son 14, father 56** — Let son = x, father = 4x. (4x + 2) + (x + 2) = 74 → 5x = 70 → x = 14.

243. **2** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 239 mod 4 = 3 → 2.

244. **70** — Order does not matter: ${}^nC_r = 8!/(4!\cdot4!) = 70$.

245. **1/18** — 2 favourable outcomes out of 36 → 2/36 = 1/18.

246. **Area 1,386 cm², circumference 132 cm** — A = πr² = 22/7 × 21² = 1,386; C = 2πr = 132.

247. **58.20** — (21×53 + 14×66) ÷ 35 = 2037 ÷ 35.

248. **40% profit** — Profit = 960. % = 960/2400 × 100 = 40%.

249. **32%** — Net = a + b − ab/100 = 20 + 15 − 300/100 = 32%.

250. **₹8,400** — $A = P(1 + R/100)^n = 40{,}000 \times (110/100)^2 = ₹48{,}400$. CI = A − P = ₹8,400.

251. **4 days** — Total work = LCM(6,12) = 12 units. A does 2/day, B does 1/day; together 3/day. Time = 12/3 = 4 days.

252. **14 2/5 s** — Relative speed = 45 km/h = 12 1/2 m/s. Distance = 100 + 80 = 180 m. Time = 14 2/5 s.

253. **Boat 8 km/h, stream 6 km/h** — Boat = (down + up)/2 = (14 + 2)/2 = 8. Stream = (down − up)/2 = 6.

254. **₹5,379** — Ratio = 30,000×12 : 60,000×8 = 3 : 4. A's share = 3/7 × 12,551 = 5,379.

255. **37 : 13** — Alligation cross: (70 − 33) : (33 − 20) = 37 : 13 = 37 : 13.

256. **Saturday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

257. **18 : 7** — Alligation cross: (90 − 54) : (54 − 40) = 36 : 14 = 18 : 7.

258. **Monday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

259. **16 : 9** — Alligation cross: (80 − 48) : (48 − 30) = 32 : 18 = 16 : 9.

260. **Tuesday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

261. **1 : 4** — Alligation cross: (70 − 64) : (64 − 40) = 6 : 24 = 1 : 4.

262. **Sunday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

263. **9 : 11** — Alligation cross: (70 − 61) : (61 − 50) = 9 : 11 = 9 : 11.

264. **Wednesday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

265. **60** — The digit is 6; tens = 10; 6 × 10 = 60.

266. **−33** — Subtracting 24 is the same as adding −24. 19 + −24 + −28 = −33.

267. **57,863** — All from 9, last from 10: 9−4=5 9−2=7 9−1=8 9−3=6, 10−7=3 → 57863.

268. **4/17** — HCF of 20 and 85 is 5. Divide both by 5.

269. **10%** — Change = 200. % change = change ÷ original × 100 = 200/2000 × 100 = 10%.

270. **9,261** — 21 × 21 × 21 = 9,261.

271. **2 × 3 × 5 × 7 × 11** — Divide repeatedly by the smallest prime that works (factor tree).

272. **1,02,861** — Write first digit 9 and last digit 1 In between, add neighbours: 9+3=12, 3+5=8, 5+1=6 Carry where a sum ≥ 10 Answer = 1,02,861

273. **8,24,824** — 1001 = 1000 + 1, so write 824 twice (padded): 824 × 1001 = 8,24,824.

274. **21,024** — Same leading part 14; last digits 4 + 6 = 10 ✓ Left: 14 × 15 = 210 Right: 4 × 6 = 24 (always two digits) Answer = 21,024

275. **1,56,021** — Working base 400 = 4 × 100. Deviations −3, −7. Left: (397 −7) × 4 = 1560. Right: −3 × −7 = 21. 1560 × 100 + 21 = 1,56,021.

276. **18,225** — Duplexes from left: 1, 6, 19, 30, 25 Combine with carries (right to left) → 18,225

277. **186,802,283** — International grouping uses threes from the right: thousand, million, billion.

278. **276** — Subtract 3,130 instead: 3,403 − 3,130 = 273; add back 3 → 276.

279. **87 1/2%** — 7/8 × 100 = 87 1/2%.

280. **10.81** — Multiply without points: 47 × 23 = 1081. Decimal places: 1 + 1 = 2, so the answer is 10.81.

281. **37** — Last digit 9 → root ends in 3 or 7. Leading part 13 lies between 3^2 and 4^2, so tens digit is 3. Check 35^2 = 1225 to choose → 37.

282. **25** — Subtract exponents: 5^2.

283. **1,810** — 5 = 10 / 2 → 362 × 10 ÷ 2 = 1,810.

284. **9,310** — Base = 100 Deviations: 95 → −5, 98 → −2 Left part (cross): 95 − 2 = 93 Right part: (−5) × (−2) = 10 Right part padded to 2 digit(s): 10 Answer = 9,310

285. **21,025** — Digits before 5: 14 14 × (one more = 15) = 210 Attach 25 → 21,025

286. **7,96,005** — Column 1 (from right): 1×5 = 5 → write 5, carry 0 Column 2 (from right): 1×5 + 3×5 = 20 → write 0, carry 2 Column 3 (from right): 1×8 + 3×5 + 9×5 = 68, + carry 2 = 70 → write 0, carry 7 Column 4 (from right): 3×8 + 9×5 = 69, + carry 7 = 76 → write 6, carry 7 Column 5 (from right): 9×8 = 72, + carry 7 = 79 → write 9, carry 7 Answer = 7,96,005

287. **Q = 430, R = 21** — Check: 22 × 430 + 21 = 9,481.

288. **Q = 69, R = 7** — Divisor 11 = 10 + 1; transposed flags: −1 Column values after applying flags: [7, -1] | [7] Resolve negatives/carries → quotient 69, remainder 7

289. **Prime** — √331 ≈ 18. No prime up to 18 divides 331, so it is prime.

290. **458** — Hundreds 400, tens 50, units 8: 400 + 50 + 8 = 458.

291. **31,814** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 8.

292. **4** — Split 5% into easy parts (10%, 5%, 1% …) or use fractions: 5% = 1/20. Result 4.

293. **52** — 16 2/3% = 1/6. 1/6 × 312 = 52.

294. **29** — Last digit 9 → cube root ends in 9 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 24 lies between 2^3 and 3^3 → tens digit 2.

295. **5** — 65,681 = 13 × 5,052 + 5.

296. **1,287** — Left: 13 − 1 = 12 Right: 99 − 12 = 87 Answer = 1,287

297. **11,440** — Base = 100 Deviations: 110 → +10, 104 → +4 Left part (cross): 110 + 4 = 114 Right part: (+10) × (+4) = 40 Right part padded to 2 digit(s): 40 Answer = 11,440

298. **224** — Same signs → positive; different signs → negative. 14 × 16 = 224.

299. **11,664** — Base 100, deviation +8 Left: 108 + 8 = 116 Right: 8^2 = 64 (needs 2 digit(s)) Answer = 11,664

300. **Q = 7,970, R = 5** — Running digit sums (quotient columns): 7, 8, 15, 18 Last running sum + last digit = remainder column 23 Adjust carries/remainder ≥ 9 → quotient 7,970, remainder 5

301. **Yes** — Rule for 8: last three digits divisible by 8. 89,312 ÷ 8 leaves remainder 0.

302. **3,138** — 197 = 200 − 3. 2,941 + 200 = 3,141, then − 3 = 3,138.

303. **≈ 49,000 (exact 53,568)** — 72 ≈ 70, 744 ≈ 700; 70 × 700 = 49,000. This is an estimate, not the exact product.

304. **1 41/132** — LCM of denominators = 132. Convert, then add numerators and simplify: 173/132.

305. **1,600** — 40 × 40 = 1600.

306. **HCF 3, LCM 120** — HCF × LCM = product: 3 × 120 = 360 = 15 × 24.

307. **≈ 12.17 (true value 12.166)** — Nearest perfect square 144 = 12^2. √n ≈ 12 + (148 − 144) ÷ (2 × 12) = 12.167.

308. **128** — (10 + 6) × 8 = 80 + 48 = 128.

309. **2,080** — Column 1 (from right): 6×0 = 0 → write 0, carry 0 Column 2 (from right): 6×8 + 2×0 = 48 → write 8, carry 4 Column 3 (from right): 2×8 = 16, + carry 4 = 20 → write 0, carry 2 Answer = 2,080

310. **10,044** — Base = 100 Deviations: 108 → +8, 93 → −7 Left part (cross): 108 − 7 = 101 Right part: (+8) × (−7) = −56 Right part is negative: borrow 1 from left → 101 × 100 − 56 Answer = 10,044

311. **2,916** — Deviation from 50: +4 Left: 25 + 4 = 29 Right: 4^2 = 16 Answer = 2,916

312. **73.46** — ÷ 50 = × 2 ÷ 100: 3,673 × 2 = 7,346, ÷ 100 = 73.46.

313. **71,159** — Write first digit 6 and last digit 9 In between, add neighbours: 6+4=10, 4+6=10, 6+9=15 Carry where a sum ≥ 10 Answer = 71,159

314. **72** — All from 9, last from 10: 9−2=7, 10−8=2 → 72.

315. **2,766** — 19 = 20 − 1. 2,747 + 20 = 2,767, then − 1 = 2,766.

316. **2,940** — 5 = 10 / 2 → 588 × 10 ÷ 2 = 2,940.

317. **2,024** — Same leading part 4; last digits 4 + 6 = 10 ✓ Left: 4 × 5 = 20 Right: 4 × 6 = 24 (always two digits) Answer = 2,024

318. **8,633** — Base = 100 Deviations: 89 → −11, 97 → −3 Left part (cross): 89 − 3 = 86 Right part: (−11) × (−3) = 33 Right part padded to 2 digit(s): 33 Answer = 8,633

319. **12** — Split 30% into easy parts (10%, 5%, 1% …) or use fractions: 30% = 3/10. Result 12.

320. **132** — 11 groups of 12 = 132.

321. **23.88** — ÷ 50 = × 2 ÷ 100: 1,194 × 2 = 2,388, ÷ 100 = 23.88.

322. **3,249** — 57 × 57 = 3249.

323. **990** — Left: 10 − 1 = 9 Right: 99 − 9 = 90 Answer = 990

324. **99** — 37 1/2% = 3/8. 3/8 × 264 = 99.

325. **1,392** — Hundreds 1300, tens 80, units 12: 1300 + 80 + 12 = 1392.

326. **1,344** — $a^2 - b^2 = (a + b)(a - b) = 56 \times 24 = 1{,}344$.

327. **50,149** — Write first digit 4 and last digit 9 In between, add neighbours: 4+5=9, 5+5=10, 5+9=14 Carry where a sum ≥ 10 Answer = 50,149

328. **6,255** — All from 9, last from 10: 9−9=0 9−3=6 9−7=2 9−4=5, 10−5=5 → 06255.

329. **9,338** — 29 = 30 − 1. 9,309 + 30 = 9,339, then − 1 = 9,338.

330. **23,225** — 25 = 100 / 4 → 929 × 100 ÷ 4 = 23,225.

331. **18,216** — Same leading part 13; last digits 2 + 8 = 10 ✓ Left: 13 × 14 = 182 Right: 2 × 8 = 16 (always two digits) Answer = 18,216

332. **8,360** — Base = 100 Deviations: 95 → −5, 88 → −12 Left part (cross): 95 − 12 = 83 Right part: (−5) × (−12) = 60 Right part padded to 2 digit(s): 60 Answer = 8,360

333. **386** — All from 9, last from 10: 9−6=3 9−1=8, 10−4=6 → 386.

334. **1,356** — 9 = 10 − 1. 1,347 + 10 = 1,357, then − 1 = 1,356.

335. **3,475** — 25 = 100 / 4 → 139 × 100 ÷ 4 = 3,475.

336. **9,52,540** — Base = 1,000 Deviations: 970 → −30, 982 → −18 Left part (cross): 970 − 18 = 952 Right part: (−30) × (−18) = 540 Right part padded to 3 digit(s): 540 Answer = 9,52,540

337. **54** — Split 45% into easy parts (10%, 5%, 1% …) or use fractions: 45% = 9/20. Result 54.

338. **24** — 2 groups of 12 = 24.

339. **1,266.2** — ÷ 5 = × 2 ÷ 10: 6,331 × 2 = 12,662, ÷ 10 = 1,266.2.

340. **225** — 15 × 15 = 225.

341. **9,108** — Left: 92 − 1 = 91 Right: 99 − 91 = 08 Answer = 9,108

342. **108** — 16 2/3% = 1/6. 1/6 × 648 = 108.

343. **935** — Hundreds 800, tens 120, units 15: 800 + 120 + 15 = 935.

344. **1,600** — $a^2 - b^2 = (a + b)(a - b) = 50 \times 32 = 1{,}600$.

345. **50,380** — Write first digit 4 and last digit 0 In between, add neighbours: 4+5=9, 5+8=13, 8+0=8 Carry where a sum ≥ 10 Answer = 50,380

346. **78,157** — All from 9, last from 10: 9−2=7 9−1=8 9−8=1 9−4=5, 10−3=7 → 78157.

347. **8,557** — 9 = 10 − 1. 8,548 + 10 = 8,558, then − 1 = 8,557.

348. **1,02,000** — 125 = 1000 / 8 → 816 × 1000 ÷ 8 = 1,02,000.

349. **1,216** — Same leading part 3; last digits 2 + 8 = 10 ✓ Left: 3 × 4 = 12 Right: 2 × 8 = 16 (always two digits) Answer = 1,216

350. **8,712** — Base = 100 Deviations: 88 → −12, 99 → −1 Left part (cross): 88 − 1 = 87 Right part: (−12) × (−1) = 12 Right part padded to 2 digit(s): 12 Answer = 8,712

351. **120** — Split 75% into easy parts (10%, 5%, 1% …) or use fractions: 75% = 3/4. Result 120.

352. **48** — 12 groups of 4 = 48.

353. **849** — All from 9, last from 10: 9−1=8 9−5=4, 10−1=9 → 849.

354. **7,755** — 9 = 10 − 1. 7,746 + 10 = 7,756, then − 1 = 7,755.

355. **2,705** — 5 = 10 / 2 → 541 × 10 ÷ 2 = 2,705.

356. **2,009** — Same leading part 4; last digits 9 + 1 = 10 ✓ Left: 4 × 5 = 20 Right: 9 × 1 = 09 (always two digits) Answer = 2,009

357. **9,108** — Base = 100 Deviations: 92 → –8, 99 → –1 Left part (cross): 92 – 1 = 91 Right part: (–8) × (–1) = 8 Right part padded to 2 digit(s): 08 Answer = 9,108

358. **360** — Split 30% into easy parts (10%, 5%, 1% …) or use fractions: 30% = 3/10. Result 360.

359. **54** — 9 groups of 6 = 54.

360. **50.6** — ÷ 50 = × 2 ÷ 100: 2,530 × 2 = 5,060, ÷ 100 = 50.6.

361. **484** — 22 × 22 = 484.

362. **3,90,609** — Left: 391 – 1 = 390 Right: 999 – 390 = 609 Answer = 3,90,609

363. **87** — 37 1/2% = 3/8. 3/8 × 232 = 87.

364. **1,288** — Hundreds 1200, tens 80, units 8: 1200 + 80 + 8 = 1288.

365. **7,623** — $a^2 - b^2 = (a + b)(a - b) = 121 \times 63 = 7{,}623$.

366. **84,777** — Write first digit 7 and last digit 7 In between, add neighbours: 7+7=14, 7+0=7, 0+7=7 Carry where a sum ≥ 10 Answer = 84,777

367. **13,225** — Digits before 5: 11 11 × (one more = 12) = 132 Attach 25 → 13,225

368. **73** — All from 9, last from 10: 9–2=7, 10–7=3 → 73.

369. **2,430** — 199 = 200 – 1. 2,231 + 200 = 2,431, then – 1 = 2,430.

370. **89,625** — 125 = 1000 / 8 → 717 × 1000 ÷ 8 = 89,625.

371. **7,209** — Same leading part 8; last digits 1 + 9 = 10 ✓ Left: 8 × 9 = 72 Right: 1 × 9 = 09 (always two digits) Answer = 7,209

372. **9,408** — Base = 100 Deviations: 96 → –4, 98 → –2 Left part (cross): 96 – 2 = 94 Right part: (–4) × (–2) = 8 Right part padded to 2 digit(s): 08 Answer = 9,408

373. **5,104** — 99 = 100 – 1. 5,005 + 100 = 5,105, then – 1 = 5,104.

374. **1,080** — 5 = 10 / 2 → 216 × 10 ÷ 2 = 1,080.

375. **13,209** — Same leading part 11; last digits 9 + 1 = 10 ✓ Left: 11 × 12 = 132 Right: 9 × 1 = 09 (always two digits) Answer = 13,209

376. **9,52,527** — Base = 1,000 Deviations: 983 → –17, 969 → –31 Left part (cross): 983 – 31 = 952 Right part: (–17) × (–31) = 527 Right part padded to 3 digit(s): 527 Answer = 9,52,527

377. **960** — Split 20% into easy parts (10%, 5%, 1% …) or use fractions: 20% = 1/5. Result 960.

378. **120** — 10 groups of 12 = 120.

379. **1,328.8** — ÷ 5 = × 2 ÷ 10: 6,644 × 2 = 13,288, ÷ 10 = 1,328.8.

380. **841** — 29 × 29 = 841.

381. **5,643** — Left: 57 – 1 = 56 Right: 99 – 56 = 43 Answer = 5,643

382. **128** — 16 2/3% = 1/6. 1/6 × 768 = 128.

383. **594** — Hundreds 500, tens 80, units 14: 500 + 80 + 14 = 594.

384. **1,716** — $a^2 - b^2 = (a + b)(a - b) = 66 \times 26 = 1{,}716$.

385. **93,632** — Write first digit 8 and last digit 2 In between, add neighbours: 8+5=13, 5+1=6, 1+2=3 Carry where a sum ≥ 10 Answer = 93,632

386. **415** — All from 9, last from 10: 9–9=0 9–5=4 9–8=1, 10–5=5 → 0415.

387. **1,622** — 29 = 30 – 1. 1,593 + 30 = 1,623, then – 1 = 1,622.

388. **38,150** — 50 = 100 / 2 → 763 × 100 ÷ 2 = 38,150.

389. **11,016** — Same leading part 10; last digits 8 + 2 = 10 ✓ Left: 10 × 11 = 110 Right: 8 × 2 = 16 (always two digits) Answer = 11,016

390. **9,118** — Base = 100 Deviations: 97 → –3, 94 → –6 Left part (cross): 97 – 6 = 91 Right part: (–3) × (–6) = 18 Right part padded to 2 digit(s): 18 Answer = 9,118

391. **8** — Split 20% into easy parts (10%, 5%, 1% …) or use fractions: 20% = 1/5. Result 8.

392. **36** — 9 groups of 4 = 36.

393. **≈ 18.33 (true value 18.330)** — Nearest perfect square 324 = 18^2. $\sqrt{n} \approx 18 + (336 - 324) \div (2 \times 18) = 18.333$.

394. **9,900** — Base = 100 Deviations: 110 → +10, 90 → −10 Left part (cross): 110 − 10 = 100 Right part: (+10) × (−10) = −100 Right part is negative: borrow 1 from left → 100 × 100 − 100 Answer = 9,900

395. **1,62,384** — Working base 400 = 4 × 100. Deviations +8, −2. Left: (408 −2) × 4 = 1624. Right: 8 × −2 = −16. 1624 × 100 + −16 = 1,62,384.

396. **58,435** — Column 1 (from right): 5×3 = 15 → write 5, carry 1 Column 2 (from right): 5×0 + 4×3 = 12, + carry 1 = 13 → write 3, carry 1 Column 3 (from right): 5×4 + 4×0 + 1×3 = 23, + carry 1 = 24 → write 4, carry 2 Column 4 (from right): 4×4 + 1×0 = 16, + carry 2 = 18 → write 8, carry 1 Column 5 (from right): 1×4 = 4, + carry 1 = 5 → write 5, carry 0 Answer = 58,435

397. **7,24,201** — Duplexes from left: 64, 80, 41, 10, 1 Combine with carries (right to left) → 7,24,201

398. **Q = 84, R = 7** — Divisor 8 = 10 − 2; write complement digits 2 Split dividend: 67 | 9 Column totals after multiplying by the complement: [6, 19] | [47] Resolve carries and remainder → quotient 84, remainder 7

399. **Q = 71, R = 3** — Divisor 13 = 10 + 3; transposed flags: −3 Column values after applying flags: [9, -25] | [81] Resolve negatives/carries → quotient 71, remainder 3

400. **0.(025641)** — Ekadhika (one more than 3) = 4 Start with 1 on the right; repeatedly multiply by 4, write the unit digit, carry the rest Recurring block (6 digits): 025641

401. **x = 5, y = 8** — Add: 54(x + y) = 702 → x + y = 13. Subtract: -10(x − y) = 30 → x − y = −3. Solve: x = 5, y = 8.

402. **x = −21** — x = (cd − ab)/(a + b − c − d) = (12 − 33)/(14 − 13).

403. **x = −4, −3, −2** — $(x + 3)^3 = x^3 + 9x^2 + 27x + 27$. The given cubic = $(x + 3)^3 - (x + 3)$. Let y = x + 3: $y^3 - y = 0 \rightarrow y = 0, \pm 1$.

404. **x = 0 or −4** — Differential: 2x + 4. Discriminant = 16 − 0 = 16, √ = 4. 2x + 4 = ±4.

405. **19 : 1** — Alligation cross: (70 − 32) : (32 − 30) = 38 : 2 = 19 : 1.

406. **Wednesday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.

407. **≈ 16.76 (true value 16.763)** — Nearest perfect square 289 = 17^2. $\sqrt{n} \approx 17 + (281 - 289) \div (2 \times 17) = 16.765$.

408. **9,682** — Base = 100 Deviations: 103 → +3, 94 → −6 Left part (cross): 103 − 6 = 97 Right part: (+3) × (−6) = −18 Right part is negative: borrow 1 from left → 97 × 100 − 18 Answer = 9,682

409. **93,324** — Working base 300 = 3 × 100. Deviations +8, +3. Left: (308 +3) × 3 = 933. Right: 8 × 3 = 24. 933 × 100 + 24 = 93,324.

410. **7,59,510** — Column 1 (from right): 0×3 = 0 → write 0, carry 0 Column 2 (from right): 0×8 + 7×3 = 21 → write 1, carry 2 Column 3 (from right): 0×7 + 7×8 + 9×3 = 83, + carry 2 = 85 → write 5, carry 8 Column 4 (from right): 7×7 + 9×8 = 121, + carry 8 = 129 → write 9, carry 12 Column 5 (from right): 9×7 = 63, + carry 12 = 75 → write 5, carry 7 Answer = 7,59,510

411. **40,000** — Duplexes from left: 4, 0, 0, 0, 0 Combine with carries (right to left) → 40,000

412. **Q = 65, R = 4** — Divisor 7 = 10 − 3; write complement digits 3 Split dividend: 45 | 9 Column totals after multiplying by the complement: [4, 17] | [60] Resolve carries and remainder → quotient 65, remainder 4

Part XIII

Solved Practice Sets

Seventy practice sets in seven categories, with ten sets in each. Every set shows a difficulty level and a suggested time. Answers and explanations follow each category.

Chapter 52: Seventy Practice Sets

Structure of every category

- Sets 1–3: **Beginner** (about 45 seconds per question)
- Sets 4–7: **Intermediate** (about 75 seconds per question)
- Sets 8–10: **Advanced** (about 2 minutes per question)

The suggested time is a target, not a rule. Accuracy comes first.

#	Category	Sets
1	Foundational arithmetic	1–10
2	Multiplication and division	11–20
3	Squares, cubes and roots	21–30
4	Algebra and factorisation	31–40
5	Percentages, ratios and averages	41–50
6	Government-exam mixed aptitude	51–60
7	Advanced mental-math challenge	61–70

After each set

Write your score, time taken, and the topic of every mistake in the tracker. Once your accuracy on a category reaches 90%, move to its next difficulty level.

Foundational arithmetic — 10 sets

✎ Set 1: Foundational arithmetic · Beginner · suggested time 6 min

413. Add mentally from left to right: 606 + 582

414. 6,784 + 99

415. 10,000 − 3,191

416. 7,784 − 129

417. Calculate 27 − 20 + 14.

418. Simplify 45/144.

419. Find 120% of 120.

420. Is 29,832 divisible by 11? Use the divisibility rule.

Answers: see the answer key (questions 413–420).

✎ Set 2: Foundational arithmetic · Beginner · suggested time 6 min

421. 2,404 + 99

422. 1,00,000 − 27,461

423. 6,926 − 889

424. Calculate 23 − (−13) + (−19).

425. Simplify 39/57.

426. Find 60% of 1,200.

427. Is 87,948 divisible by 8? Use the divisibility rule.

428. Add mentally from left to right: 719 + 304

Answers: see the answer key (questions 421–428).

✎ Set 3: Foundational arithmetic · Beginner · suggested time 6 min

429. 100 – 82

430. 2,356 – 1,079

431. Calculate 23 – (–28) + (–24).

432. Simplify 8/52.

433. Find 25% of 240.

434. Is 93,705 divisible by 3? Use the divisibility rule.

435. Add mentally from left to right: 209 + 205

436. 9,121 + 9

Answers: see the answer key (questions 429–436).

✎ Set 4: Foundational arithmetic - Intermediate - suggested time 10 min

437. Find the HCF and LCM of 24 and 96.

438. Estimate 90 × 786 by rounding (approximate answer). *(approximate)*

439. Find the remainder when 98,032 is divided by 13.

440. $2^5 \times 2^1$ (give the value)

441. Add: 9,090 + 763 + 3,646 + 9,251 + 2,328

442. 3/2 − 4/3

443. 3.2 × 3.8

444. Find the HCF and LCM of 12 and 16.

Answers: see the answer key (questions 437–444).

✎ Set 5: Foundational arithmetic - Intermediate - suggested time 10 min

445. Estimate 70 × 862 by rounding (approximate answer). *(approximate)*

446. Find the remainder when 21,769 is divided by 11.

447. $10^3 \times 10^1$ (give the value)

448. Add: 1,883 + 195 + 551 + 1,739 + 6,212 + 3,640

449. 9/10 – 2

450. 7.1 × 0.84

451. Find the HCF and LCM of 27 and 81.

452. Estimate 96 × 247 by rounding (approximate answer). *(approximate)*

Answers: see the answer key (questions 445–452).

✎ Set 6: Foundational arithmetic - Intermediate - suggested time 10 min

453. Find the remainder when 82,183 is divided by 7.

454. $2^6 \times 2^2$ (give the value)

455. Add: 6,326 + 2,451 + 2,948 + 2,879 + 577

456. 8/9 + 8/5

457. 5.1 × 1.2

458. Find the HCF and LCM of 24 and 8.

459. Estimate 93 × 348 by rounding (approximate answer). *(approximate)*

460. Find the remainder when 49,409 is divided by 9.

Answers: see the answer key (questions 453–460).

✎ Set 7: Foundational arithmetic - Intermediate - suggested time 10 min

461. $3^5 \times 3^1$ (give the value)

462. Add: 377 + 7,396 + 3,801 + 7,666 + 6,443 + 2,011

463. 7/4 – 9/7

464. 6.2 × 6.2

465. Find the HCF and LCM of 42 and 56.

466. Estimate 33 × 946 by rounding (approximate answer). *(approximate)*

467. Find the remainder when 42,508 is divided by 9.

468. $5^7 \div 5^4$

Answers: see the answer key (questions 461–468).

✎ Set 8: Foundational arithmetic · Advanced · suggested time 16 min

469. Add: 8,114 + 6,097 + 5,186 + 3,753

470. 1/2 + 1/10

471. 9.8 × 0.51

472. Find the HCF and LCM of 64 and 40.

473. Estimate 60 × 691 by rounding (approximate answer). *(approximate)*

474. Find the remainder when 82,837 is divided by 7.

475. $10^5 \div 10^3$

476. Add: 8,895 + 7,190 + 2,553 + 2,768 + 9,401 + 2,214

Answers: see the answer key (questions 469–476).

✎ Set 9: Foundational arithmetic · Advanced · suggested time 16 min

477. 1/4 – 1/2

478. 1.4 × 0.18

479. Find the HCF and LCM of 112 and 70.

480. Estimate 45 × 280 by rounding (approximate answer). *(approximate)*

481. Find the remainder when 22,168 is divided by 7.

482. 10^{-3} as a fraction

483. Add: 9,252 + 4,967 + 813 + 7,833 + 3,930 + 2,017

484. 4/3 – 5/9

Answers: see the answer key (questions 477–484).

✎ Set 10: Foundational arithmetic - Advanced - suggested time 16 min

485. 6.1 × 8.8

486. Find the HCF and LCM of 96 and 40.

487. Estimate 40 × 811 by rounding (approximate answer). *(approximate)*

488. Find the remainder when 85,396 is divided by 9.

489. $2^2 \times 2^4$ (give the value)

490. Add: 5,585 + 9,148 + 1,439 + 3,180 + 1,062 + 429

491. 1 − 1/9

492. 6.4 × 0.95

Answers: see the answer key (questions 485–492).

Answers and explanations — Foundational arithmetic

413. **1,188** — Hundreds 1100, tens 80, units 8: 1100 + 80 + 8 = 1188.

414. **6,883** — 99 = 100 − 1. 6,784 + 100 = 6,884, then − 1 = 6,883.

415. **6,809** — All from 9, last from 10: 9−3=6 9−1=8 9−9=0, 10−1=9 → 6809.

416. **7,655** — Subtract 130 instead: 7,784 − 130 = 7,654; add back 1 → 7,655.

417. **21** — Subtracting 20 is the same as adding −20. 27 + −20 + 14 = 21.

418. **5/16** — HCF of 45 and 144 is 9. Divide both by 9.

419. **144** — Split 120% into easy parts (10%, 5%, 1% …) or use fractions: 120% = 6/5. Result 144.

420. **Yes** — Rule for 11: (sum of odd-place digits − sum of even-place digits) divisible by 11. 29,832 ÷ 11 leaves remainder 0.

421. **2,503** — 99 = 100 − 1. 2,404 + 100 = 2,504, then − 1 = 2,503.

422. **72,539** — All from 9, last from 10: 9−2=7 9−7=2 9−4=5 9−6=3, 10−1=9 → 72539.

423. **6,037** — Subtract 890 instead: 6,926 − 890 = 6,036; add back 1 → 6,037.

424. **17** — Subtracting (−13) is the same as adding 13. 23 + 13 + −19 = 17.

425. **13/19** — HCF of 39 and 57 is 3. Divide both by 3.

426. **720** — Split 60% into easy parts (10%, 5%, 1% …) or use fractions: 60% = 3/5. Result 720.

427. **No** — Rule for 8: last three digits divisible by 8. 87,948 ÷ 8 leaves remainder 4.

428. **1,023** — Hundreds 1000, tens 10, units 13: 1000 + 10 + 13 = 1023.

429. **18** — All from 9, last from 10: 9−8=1, 10−2=8 → 18.

430. **1,277** — Subtract 1,080 instead: 2,356 − 1,080 = 1,276; add back 1 → 1,277.

431. **27** — Subtracting (−28) is the same as adding 28. 23 + 28 + −24 = 27.

432. **2/13** — HCF of 8 and 52 is 4. Divide both by 4.

433. **60** — Split 25% into easy parts (10%, 5%, 1% …) or use fractions: 25% = 1/4. Result 60.

434. **Yes** — Rule for 3: digit sum divisible by 3. 93,705 ÷ 3 leaves remainder 0.

435. **414** — Hundreds 400, tens 0, units 14: 400 + 0 + 14 = 414.

436. **9,130** — 9 = 10 − 1. 9,121 + 10 = 9,131, then − 1 = 9,130.

437. **HCF 24, LCM 96** — HCF × LCM = product: 24 × 96 = 2304 = 24 × 96.

438. **≈ 72,000 (exact 70,740)** — 90 ≈ 90, 786 ≈ 800; 90 × 800 = 72,000. This is an estimate, not the exact product.

439. **12** — 98,032 = 13 × 7,540 + 12.

440. **64** — Add exponents: $2^6 = 64$.

441. **25,078** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 4.

442. **1/6** — LCM of denominators = 6. Convert, then subtract numerators and simplify: 1/6.

443. **12.16** — Multiply without points: 32 × 38 = 1216. Decimal places: 1 + 1 = 2, so the answer is 12.16.

444. **HCF 4, LCM 48** — HCF × LCM = product: 4 × 48 = 192 = 12 × 16.

445. **≈ 63,000 (exact 60,340)** — 70 ≈ 70, 862 ≈ 900; 70 × 900 = 63,000. This is an estimate, not the exact product.

446. **0** — 21,769 = 11 × 1,979 + 0.

447. **10,000** — Add exponents: $10^4 = 10{,}000$.

448. **14,220** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 9.

449. **−1 1/10** — LCM of denominators = 10. Convert, then subtract numerators and simplify: −11/10.

450. **5.964** — Multiply without points: 71 × 84 = 5964. Decimal places: 1 + 2 = 3, so the answer is 5.964.

451. **HCF 27, LCM 81** — HCF × LCM = product: 27 × 81 = 2187 = 27 × 81.

452. **≈ 25,000 (exact 23,712)** — 96 ≈ 100, 247 ≈ 250; 100 × 250 = 25,000. This is an estimate, not the exact product.

453. **3** — 82,183 = 7 × 11,740 + 3.

454. **256** — Add exponents: $2^8 = 256$.

455. **15,181** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 7.

456. **2 22/45** — LCM of denominators = 45. Convert, then add numerators and simplify: 112/45.

457. **6.12** — Multiply without points: 51 × 12 = 612. Decimal places: 1 + 1 = 2, so the answer is 6.12.

458. **HCF 8, LCM 24** — HCF × LCM = product: 8 × 24 = 192 = 24 × 8.

459. **≈ 31,500 (exact 32,364)** — 93 ≈ 90, 348 ≈ 350; 90 × 350 = 31,500. This is an estimate, not the exact product.

460. **8** — 49,409 = 9 × 5,489 + 8. Shortcut: digit sum mod 9.

461. **729** — Add exponents: $3^6 = 729$.

462. **27,694** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 1.

463. **13/28** — LCM of denominators = 28. Convert, then subtract numerators and simplify: 13/28.

464. **38.44** — Multiply without points: 62 × 62 = 3844. Decimal places: 1 + 1 = 2, so the answer is 38.44.

465. **HCF 14, LCM 168** — HCF × LCM = product: 14 × 168 = 2352 = 42 × 56.

466. **≈ 27,000 (exact 31,218)** — 33 ≈ 30, 946 ≈ 900; 30 × 900 = 27,000. This is an estimate, not the exact product.

467. **1** — 42,508 = 9 × 4,723 + 1. Shortcut: digit sum mod 9.

468. **125** — Subtract exponents: 5^3.

469. **23,150** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 2.

470. **3/5** — LCM of denominators = 10. Convert, then add numerators and simplify: 3/5.

471. **4.998** — Multiply without points: 98 × 51 = 4998. Decimal places: 1 + 2 = 3, so the answer is 4.998.

472. **HCF 8, LCM 320** — HCF × LCM = product: 8 × 320 = 2560 = 64 × 40.

473. **≈ 42,000 (exact 41,460)** — 60 ≈ 60, 691 ≈ 700; 60 × 700 = 42,000. This is an estimate, not the exact product.

474. **6** — 82,837 = 7 × 11,833 + 6.

475. **100** — Subtract exponents: 10^2.

476. **33,021** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 9.

477. **−1/4** — LCM of denominators = 4. Convert, then subtract numerators and simplify: −1/4.

478. **0.252** — Multiply without points: 14 × 18 = 252. Decimal places: 1 + 2 = 3, so the answer is 0.252.

479. **HCF 14, LCM 560** — HCF × LCM = product: 14 × 560 = 7840 = 112 × 70.

480. **≈ 11,200 (exact 12,600)** — 45 ≈ 40, 280 ≈ 280; 40 × 280 = 11,200. This is an estimate, not the exact product.

481. **6** — 22,168 = 7 × 3,166 + 6.

482. **1/1,000** — A negative exponent means reciprocal: $1/10^3$.

483. **28,812** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 3.

484. **7/9** — LCM of denominators = 9. Convert, then subtract numerators and simplify: 7/9.

485. **53.68** — Multiply without points: 61 × 88 = 5368. Decimal places: 1 + 1 = 2, so the answer is 53.68.

486. **HCF 8, LCM 480** — HCF × LCM = product: 8 × 480 = 3840 = 96 × 40.

487. **≈ 32,000 (exact 32,440)** — 40 ≈ 40, 811 ≈ 800; 40 × 800 = 32,000. This is an estimate, not the exact product.

488. **4** — 85,396 = 9 × 9,488 + 4. Shortcut: digit sum mod 9.

489. **64** — Add exponents: $2^6 = 64$.

490. **20,843** — Group friendly pairs, or add column-wise with running totals. Digit-sum check: digital roots give 8.

491. **8/9** — LCM of denominators = 9. Convert, then subtract numerators and simplify: 8/9.

492. **6.08** — Multiply without points: 64 × 95 = 6080. Decimal places: 1 + 2 = 3, so the answer is 6.08.

Multiplication and division — 10 sets

✎ Set 11: Multiplication and division · Beginner · suggested time 6 min

493. 3,108 × 11

494. 329 × 25

495. 561 × 999

496. 4,206 ÷ 6

497. 2,212 ÷ 50 (as a decimal)

498. 91,438 ÷ 9 (quotient and remainder)

499. 1,609 × 11

500. 509 × 25

Answers: see the answer key (questions 493–500).

✎ Set 12: Multiplication and division · Beginner · suggested time 6 min

501. 237 × 50

502. 54 × 99

503. 2,944 ÷ 19

504. 7,000 ÷ 5 (as a decimal)

505. 92,998 ÷ 9 (quotient and remainder)

506. 5,920 × 11

507. 170 × 125

508. 1,474 × 9,999

Answers: see the answer key (questions 501–508).

✎ Set 13: Multiplication and division · Beginner · suggested time 6 min

509. 7,840 × 9,999

510. 7,355 ÷ 14

511. 9,265 ÷ 25 (as a decimal)

512. 55,736 ÷ 9 (quotient and remainder)

513. 8,326 × 11

514. 684 × 5

515. 181 × 999

516. 2,618 ÷ 20

Answers: see the answer key (questions 509–516).

✎ Set 14: Multiplication and division - Intermediate - suggested time 10 min

517. 152 × 158

518. 584 × 1,001

519. 96 × 92 (Nikhilam, base 100)

520. 104 × 106 (base 100)

521. 39 × 93 (vertically and crosswise)

522. 767 × 1,001

523. 96 × 95 (Nikhilam, base 100)

524. 1,037 × 1,017 (base 1,000)

Answers: see the answer key (questions 517–524).

✎ Set 15: Multiplication and division - Intermediate - suggested time 10 min

525. 837 × 1,001

526. 970 × 965 (Nikhilam, base 1,000)

527. 106 × 115 (base 100)

528. 31 × 74 (vertically and crosswise)

529. 107 × 103

530. 573 × 1,001

531. 91 × 90 (Nikhilam, base 100)

532. 105 × 102 (base 100)

Answers: see the answer key (questions 525–532).

✎ Set 16: Multiplication and division - Intermediate - suggested time 10 min

533. 90 × 92 (Nikhilam, base 100)

534. 101 × 111 (base 100)

535. 44 × 75 (vertically and crosswise)

536. 106 × 104

537. 962 × 1,001

538. 98 × 95 (Nikhilam, base 100)

539. 1,025 × 1,029 (base 1,000)

540. 96 × 65 (vertically and crosswise)

Answers: see the answer key (questions 533–540).

✎ Set 17: Multiplication and division - Intermediate - suggested time 10 min

541. 101 × 109 (base 100)

542. 59 × 59 (vertically and crosswise)

543. 35 × 35

544. 94 × 101

545. 1,003 × 1,039 (base 1,000)

546. 20 × 56 (vertically and crosswise)

547. 43 × 47

548. 31 × 101

Answers: see the answer key (questions 541–548).

✎ Set 18: Multiplication and division - Advanced - suggested time 16 min

549. 4,044 ÷ 88

550. 53,193 ÷ 1012

551. 304 × 294 (working base 300)

552. 105 × 88

553. 935 × 844

554. 3,830 ÷ 88

555. 9,871 ÷ 112

556. 51 × 58 (working base 50)

Answers: see the answer key (questions 549–556).

✎ Set 19: Multiplication and division - Advanced - suggested time 16 min

557. 488 ÷ 11

558. 207 × 207 (working base 200)

559. 110 × 98

560. 704 × 150

561. 14,917 ÷ 997

562. 5,614 ÷ 113

563. 296 × 308 (working base 300)

564. 110 × 92

Answers: see the answer key (questions 557–564).

✎ Set 20: Multiplication and division · Advanced · suggested time 16 min

565. 45 × 52 (working base 50)

566. 106 × 93

567. 790 × 934

568. 2,576 ÷ 88

569. 876 ÷ 14

570. 45 × 51 (working base 50)

571. 111 × 96

572. 331 × 153

Answers: see the answer key (questions 565–572).

Answers and explanations — Multiplication and division

493. **34,188** — Write first digit 3 and last digit 8 In between, add neighbours: 3+1=4, 1+0=1, 0+8=8 Carry where a sum ≥ 10 Answer = 34,188

494. **8,225** — 25 = 100 / 4 → 329 × 100 ÷ 4 = 8,225.

495. **5,60,439** — Left: 561 − 1 = 560 Right: 999 − 560 = 439 Answer = 5,60,439

496. **Q = 701, R = 0** — Check: 6 × 701 + 0 = 4,206.

497. **44.24** — ÷ 50 = × 2 ÷ 100: 2,212 × 2 = 4,424, ÷ 100 = 44.24.

498. **Q = 10,159, R = 7** — Running digit sums (quotient columns): 9, 10, 14, 17 Last running sum + last digit = remainder column 25 Adjust carries/remainder ≥ 9 → quotient 10,159, remainder 7

499. **17,699** — Write first digit 1 and last digit 9 In between, add neighbours: 1+6=7, 6+0=6, 0+9=9 Carry where a sum ≥ 10 Answer = 17,699

500. **12,725** — 25 = 100 / 4 → 509 × 100 ÷ 4 = 12,725.

501. **11,850** — 50 = 100 / 2 → 237 × 100 ÷ 2 = 11,850.

502. **5,346** — Left: 54 − 1 = 53 Right: 99 − 53 = 46 Answer = 5,346

503. **Q = 154, R = 18** — Check: 19 × 154 + 18 = 2,944.

504. **1,400** — ÷ 5 = × 2 ÷ 10: 7,000 × 2 = 14,000, ÷ 10 = 1,400.

505. **Q = 10,333, R = 1** — Running digit sums (quotient columns): 9, 11, 20, 29 Last running sum + last digit = remainder column 37 Adjust carries/remainder ≥ 9 → quotient 10,333, remainder 1

506. **65,120** — Write first digit 5 and last digit 0 In between, add neighbours: 5+9=14, 9+2=11, 2+0=2 Carry where a sum ≥ 10 Answer = 65,120

507. **21,250** — 125 = 1000 / 8 → 170 × 1000 ÷ 8 = 21,250.

508. **1,47,38,526** — Left: 1474 − 1 = 1473 Right: 9,999 − 1473 = 8526 Answer = 1,47,38,526

509. **7,83,92,160** — Left: 7840 − 1 = 7839 Right: 9,999 − 7839 = 2160 Answer = 7,83,92,160

510. **Q = 525, R = 5** — Check: 14 × 525 + 5 = 7,355.

511. **370.6** — ÷ 25 = × 4 ÷ 100: 9,265 × 4 = 37,060, ÷ 100 = 370.6.

512. **Q = 6,192, R = 8** — Running digit sums (quotient columns): 5, 10, 17, 20 Last running sum + last digit = remainder column 26 Adjust carries/remainder ≥ 9 → quotient 6,192, remainder 8

513. **91,586** — Write first digit 8 and last digit 6 In between, add neighbours: 8+3=11, 3+2=5, 2+6=8 Carry where a sum ≥ 10 Answer = 91,586

514. **3,420** — 5 = 10 / 2 → 684 × 10 ÷ 2 = 3,420.

515. **1,80,819** — Left: 181 − 1 = 180 Right: 999 − 180 = 819 Answer = 1,80,819

516. **Q = 130, R = 18** — Check: 20 × 130 + 18 = 2,618.

517. **24,016** — Same leading part 15; last digits 2 + 8 = 10 ✓ Left: 15 × 16 = 240 Right: 2 × 8 = 16 (always two digits) Answer = 24,016

518. **5,84,584** — 1001 = 1000 + 1, so write 584 twice (padded): 584 × 1001 = 5,84,584.

519. **8,832** — Base = 100 Deviations: 96 → −4, 92 → −8 Left part (cross): 96 − 8 = 88 Right part: (−4) × (−8) = 32 Right part padded to 2 digit(s): 32 Answer = 8,832

520. **11,024** — Base = 100 Deviations: 104 → +4, 106 → +6 Left part (cross): 104 + 6 = 110 Right part: (+4) × (+6) = 24 Right part padded to 2 digit(s): 24 Answer = 11,024

521. **3,627** — Column 1 (from right): 9×3 = 27 → write 7, carry 2 Column 2 (from right): 9×9 + 3×3 = 90, + carry 2 = 92 → write 2, carry 9 Column 3 (from right): 3×9 = 27, + carry 9 = 36 → write 6, carry 3 Answer = 3,627

522. **7,67,767** — 1001 = 1000 + 1, so write 767 twice (padded): 767 × 1001 = 7,67,767.

523. **9,120** — Base = 100 Deviations: 96 → −4, 95 → −5 Left part (cross): 96 − 5 = 91 Right part: (−4) × (−5) = 20 Right part padded to 2 digit(s): 20 Answer = 9,120

524. **10,54,629** — Base = 1,000 Deviations: 1037 → +37, 1017 → +17 Left part (cross): 1037 + 17 = 1054 Right part: (+37) × (+17) = 629 Right part padded to 3 digit(s): 629 Answer = 10,54,629

525. **8,37,837** — 1001 = 1000 + 1, so write 837 twice (padded): 837 × 1001 = 8,37,837.

526. **9,36,050** — Base = 1,000 Deviations: 970 → −30, 965 → −35 Left part (cross): 970 − 35 = 935 Right part: (−30) × (−35) = 1050 Right part has more than 3 digit(s): carry 1 to the left Answer = 9,36,050

527. **12,190** — Base = 100 Deviations: 106 → +6, 115 → +15 Left part (cross): 106 + 15 = 121 Right part: (+6) × (+15) = 90 Right part padded to 2 digit(s): 90 Answer = 12,190

528. **2,294** — Column 1 (from right): 1×4 = 4 → write 4, carry 0 Column 2 (from right): 1×7 + 3×4 = 19 → write 9, carry 1 Column 3 (from right): 3×7 = 21, + carry 1 = 22 → write 2, carry 2 Answer = 2,294

529. **11,021** — Same leading part 10; last digits 7 + 3 = 10 ✓ Left: 10 × 11 = 110 Right: 7 × 3 = 21 (always two digits) Answer = 11,021

530. **5,73,573** — 1001 = 1000 + 1, so write 573 twice (padded): 573 × 1001 = 5,73,573.

531. **8,190** — Base = 100 Deviations: 91 → −9, 90 → −10 Left part (cross): 91 − 10 = 81 Right part: (−9) × (−10) = 90 Right part padded to 2 digit(s): 90 Answer = 8,190

532. **10,710** — Base = 100 Deviations: 105 → +5, 102 → +2 Left part (cross): 105 + 2 = 107 Right part: (+5) × (+2) = 10 Right part padded to 2 digit(s): 10 Answer = 10,710

533. **8,280** — Base = 100 Deviations: 90 → −10, 92 → −8 Left part (cross): 90 − 8 = 82 Right part: (−10) × (−8) = 80 Right part padded to 2 digit(s): 80 Answer = 8,280

534. **11,211** — Base = 100 Deviations: 101 → +1, 111 → +11 Left part (cross): 101 + 11 = 112 Right part: (+1) × (+11) = 11 Right part padded to 2 digit(s): 11 Answer = 11,211

535. **3,300** — Column 1 (from right): 4×5 = 20 → write 0, carry 2 Column 2 (from right): 4×7 + 4×5 = 48, + carry 2 = 50 → write 0, carry 5 Column 3 (from right): 4×7 = 28, + carry 5 = 33 → write 3, carry 3 Answer = 3,300

536. **11,024** — Same leading part 10; last digits 6 + 4 = 10 ✓ Left: 10 × 11 = 110 Right: 6 × 4 = 24 (always two digits) Answer = 11,024

537. **9,62,962** — 1001 = 1000 + 1, so write 962 twice (padded): 962 × 1001 = 9,62,962.

538. **9,310** — Base = 100 Deviations: 98 → −2, 95 → −5 Left part (cross): 98 − 5 = 93 Right part: (−2) × (−5) = 10 Right part padded to 2 digit(s): 10 Answer = 9,310

539. **10,54,725** — Base = 1,000 Deviations: 1025 → +25, 1029 → +29 Left part (cross): 1025 + 29 = 1054 Right part: (+25) × (+29) = 725 Right part padded to 3 digit(s): 725 Answer = 10,54,725

540. **6,240** — Column 1 (from right): 6×5 = 30 → write 0, carry 3 Column 2 (from right): 6×6 + 9×5 = 81, + carry 3 = 84 → write 4, carry 8 Column 3 (from right): 9×6 = 54, + carry 8 = 62 → write 2, carry 6 Answer = 6,240

541. **11,009** — Base = 100 Deviations: 101 → +1, 109 → +9 Left part (cross): 101 + 9 = 110 Right part: (+1) × (+9) = 9 Right part padded to 2 digit(s): 09 Answer = 11,009

542. **3,481** — Column 1 (from right): 9×9 = 81 → write 1, carry 8 Column 2 (from right): 9×5 + 5×9 = 90, + carry 8 = 98 → write 8, carry 9 Column 3 (from right): 5×5 = 25, + carry 9 = 34 → write 4, carry 3 Answer = 3,481

543. **1,225** — Same leading part 3; last digits 5 + 5 = 10 ✓ Left: 3 × 4 = 12 Right: 5 × 5 = 25 (always two digits) Answer = 1,225

544. **9,494** — 101 = 100 + 1, so write 94 twice (padded): 94 × 101 = 9,494.

545. **10,42,117** — Base = 1,000 Deviations: 1003 → +3, 1039 → +39 Left part (cross): 1003 + 39 = 1042 Right part: (+3) × (+39) = 117 Right part padded to 3 digit(s): 117 Answer = 10,42,117

546. **1,120** — Column 1 (from right): 0×6 = 0 → write 0, carry 0 Column 2 (from right): 0×5 + 2×6 = 12 → write 2, carry 1 Column 3 (from right): 2×5 = 10, + carry 1 = 11 → write 1, carry 1 Answer = 1,120

547. **2,021** — Same leading part 4; last digits 3 + 7 = 10 ✓ Left: 4 × 5 = 20 Right: 3 × 7 = 21 (always two digits) Answer = 2,021

548. **3,131** — 101 = 100 + 1, so write 31 twice (padded): 31 × 101 = 3,131.

549. **Q = 45, R = 84** — Divisor 88 = 100 − 12; write complement digits 12 Split dividend: 40 | 44 Column totals after multiplying by the complement: [4, 4] | [16, 12] Resolve carries and remainder → quotient 45, remainder 84

550. **Q = 52, R = 569** — Divisor 1012 = 1,000 + 12; transposed flags: −0, −1, −2 Column values after applying flags: [5, 3] | [-4, -4, -3] Resolve negatives/carries → quotient 52, remainder 569

551. **89,376** — Working base 300 = 3 × 100. Deviations +4, −6. Left: (304 −6) × 3 = 894. Right: 4 × −6 = −24. 894 × 100 + −24 = 89,376.

552. **9,240** — Base = 100 Deviations: 105 → +5, 88 → −12 Left part (cross): 105 − 12 = 93 Right part: (+5) × (−12) = −60 Right part is negative: borrow 1 from left → 93 × 100 − 60 Answer = 9,240

553. **7,89,140** — Column 1 (from right): 5×4 = 20 → write 0, carry 2 Column 2 (from right): 5×4 + 3×4 = 32, + carry 2 = 34 → write 4, carry 3 Column 3 (from right): 5×8 + 3×4 + 9×4 = 88, + carry 3 = 91 → write 1, carry 9 Column 4 (from right): 3×8 + 9×4 = 60, + carry 9 = 69 → write 9, carry 6 Column 5 (from right): 9×8 = 72, + carry 6 = 78 → write 8, carry 7 Answer = 7,89,140

554. **Q = 43, R = 46** — Divisor 88 = 100 − 12; write complement digits 12 Split dividend: 38 | 30 Column totals after multiplying by the complement: [3, 11] | [20, 22] Resolve carries and remainder → quotient 43, remainder 46

555. **Q = 88, R = 15** — Divisor 112 = 100 + 12; transposed flags: −1, −2 Column values after applying flags: [9, -1] | [-10, 3] Resolve negatives/carries → quotient 88, remainder 15

556. **2,958** — Working base 50 = 5 × 10. Deviations +1, +8. Left: (51 +8) × 5 = 295. Right: 1 × 8 = 8. 295 × 10 + 8 = 2,958.

557. **Q = 44, R = 4** — Divisor 11 = 10 + 1; transposed flags: −1 Column values after applying flags: [4, 4] | [4] Resolve negatives/carries → quotient 44, remainder 4

558. **42,849** — Working base 200 = 2 × 100. Deviations +7, +7. Left: (207 +7) × 2 = 428. Right: 7 × 7 = 49. 428 × 100 + 49 = 42,849.

559. **10,780** — Base = 100 Deviations: 110 → +10, 98 → −2 Left part (cross): 110 − 2 = 108 Right part: (+10) × (−2) = −20 Right part is negative: borrow 1 from left → 108 × 100 − 20 Answer = 10,780

560. **1,05,600** — Column 1 (from right): 4×0 = 0 → write 0, carry 0 Column 2 (from right): 4×5 + 0×0 = 20 → write 0, carry 2 Column 3 (from right): 4×1 + 0×5 + 7×0 = 4, + carry 2 = 6 → write 6, carry 0 Column 4 (from right): 0×1 + 7×5 = 35 → write 5, carry 3 Column 5 (from right): 7×1 = 7, + carry 3 = 10 → write 0, carry 1 Answer = 1,05,600

561. **Q = 14, R = 959** — Divisor 997 = 1,000 − 3; write complement digits 003 Split dividend: 14 | 917 Column totals after multiplying by the complement: [1, 4] | [9, 4, 19] Resolve carries and remainder → quotient 14, remainder 959

562. **Q = 49, R = 77** — Divisor 113 = 100 + 13; transposed flags: −1, −3 Column values after applying flags: [5, 1] | [-15, 1] Resolve negatives/carries → quotient 49, remainder 77

563. **91,168** — Working base 300 = 3 × 100. Deviations −4, +8. Left: (296 +8) × 3 = 912. Right: −4 × 8 = −32. 912 × 100 + −32 = 91,168.

564. **10,120** — Base = 100 Deviations: 110 → +10, 92 → −8 Left part (cross): 110 − 8 = 102 Right part: (+10) × (−8) = −80 Right part is negative: borrow 1 from left → 102 × 100 − 80 Answer = 10,120

565. **2,340** — Working base 50 = 5 × 10. Deviations −5, +2. Left: (45 +2) × 5 = 235. Right: −5 × 2 = −10. 235 × 10 + −10 = 2,340.

566. **9,858** — Base = 100 Deviations: 106 → +6, 93 → −7 Left part (cross): 106 − 7 = 99 Right part: (+6) × (−7) = −42 Right part is negative: borrow 1 from left → 99 × 100 − 42 Answer = 9,858

567. **7,37,860** — Column 1 (from right): 0×4 = 0 → write 0, carry 0 Column 2 (from right): 0×3 + 9×4 = 36 → write 6, carry 3 Column 3 (from right): 0×9 + 9×3 + 7×4 = 55, + carry 3 = 58 → write 8, carry 5 Column 4 (from right): 9×9 + 7×3 = 102, + carry 5 = 107 → write 7, carry 10 Column 5 (from right): 7×9 = 63, + carry 10 = 73 → write 3, carry 7 Answer = 7,37,860

568. **Q = 29, R = 24** — Divisor 88 = 100 − 12; write complement digits 12 Split dividend: 25 | 76 Column totals after multiplying by the complement: [2, 7] | [18, 20] Resolve carries and remainder → quotient 29, remainder 24

569. **Q = 62, R = 8** — Divisor 14 = 10 + 4; transposed flags: −4 Column values after applying flags: [8, -25] | [106] Resolve negatives/carries → quotient 62, remainder 8

570. **2,295** — Working base 50 = 5 × 10. Deviations −5, +1. Left: (45 +1) × 5 = 230. Right: −5 × 1 = −5. 230 × 10 + −5 = 2,295.

571. **10,656** — Base = 100 Deviations: 111 → +11, 96 → −4 Left part (cross): 111 − 4 = 107 Right part: (+11) × (−4) = −44 Right part is negative: borrow 1 from left → 107 × 100 − 44 Answer = 10,656

572. **50,643** — Column 1 (from right): 1×3 = 3 → write 3, carry 0 Column 2 (from right): 1×5 + 3×3 = 14 → write 4, carry 1 Column 3 (from right): 1×1 + 3×5 + 3×3 = 25, + carry 1 = 26 → write 6, carry 2 Column 4 (from right): 3×1 + 3×5 = 18, + carry 2 = 20 → write 0, carry 2 Column 5 (from right): 3×1 = 3, + carry 2 = 5 → write 5, carry 0 Answer = 50,643

Squares, cubes and roots — 10 sets

✎ Set 21: Squares, cubes and roots · Beginner · suggested time 6 min

573. 45^2

574. 41^2

575. 35^2

576. 125^2

577. 195^2

578. 37^2

579. 75^2

580. 155^2

Answers: see the answer key (questions 573–580).

✎ Set 22: Squares, cubes and roots · Beginner · suggested time 6 min

581. 19^2

582. 28^2

583. 105^2

584. 16^2

585. 175^2

586. 17^2

587. 95^2

588. 30^2

Answers: see the answer key (questions 581–588).

✎ Set 23: Squares, cubes and roots · Beginner · suggested time 6 min

589. 185^2

590. 39^2

591. 23^2

592. 34^2

593. 58^2

594. 38^2

595. 51^2

596. 12^2

Answers: see the answer key (questions 589–596).

✎ Set 24: Squares, cubes and roots · Intermediate · suggested time 10 min

597. $\sqrt{5,041}$

598. $\sqrt[3]{1,25,000}$

599. Use $(a \pm b)^2$ to find 97^2

600. 5^3

601. 107^2

602. $\sqrt{1,681}$

603. $\sqrt[3]{35,937}$

604. Use $(a \pm b)^2$ to find 63^2

Answers: see the answer key (questions 597–604).

✎ Set 25: Squares, cubes and roots · Intermediate · suggested time 10 min

605. Use $(a \pm b)^2$ to find 301^2

606. 114^2

607. 46^2

608. $\sqrt{2,209}$

609. $\sqrt[3]{2,87,496}$

610. Use $(a \pm b)^2$ to find 69^2

611. 24^3

612. 960^2

Answers: see the answer key (questions 605–612).

✎ Set 26: Squares, cubes and roots · Intermediate · suggested time 10 min

613. Use $(a \pm b)^2$ to find 82^2

614. 978^2

615. $\sqrt{1,156}$

616. $\sqrt[3]{3,57,911}$

617. Use $(a \pm b)^2$ to find 79^2

618. 996^2

619. $\sqrt{3,481}$

620. $\sqrt[3]{12,167}$

Answers: see the answer key (questions 613–620).

✎ Set 27: Squares, cubes and roots · Intermediate · suggested time 10 min

621. 93^2

622. $\sqrt{676}$

623. $\sqrt[3]{1,10,592}$

624. Use $(a \pm b)^2$ to find 302^2

625. $\sqrt[3]{3,43,000}$

626. Use $(a \pm b)^2$ to find 21^2

627. 10^3

628. 103^2

Answers: see the answer key (questions 621–628).

✎ Set 28: Squares, cubes and roots - Advanced - suggested time 16 min

629. Estimate √286 to 2 decimal places (approximate). *(approximate)*

630. 400^2 (duplex method)

631. Estimate √165 to 2 decimal places (approximate). *(approximate)*

632. 964^2 (duplex method)

633. Estimate √376 to 2 decimal places (approximate). *(approximate)*

634. 730^2 (duplex method)

635. Estimate √90 to 2 decimal places (approximate). *(approximate)*

636. 109^2 (duplex method)

Answers: see the answer key (questions 629–636).

✎ Set 29: Squares, cubes and roots - Advanced - suggested time 16 min

637. 532^2 (duplex method)

638. Estimate $\sqrt{77}$ to 2 decimal places (approximate). *(approximate)*

639. 396^2 (duplex method)

640. 331^2 (duplex method)

641. Estimate $\sqrt{274}$ to 2 decimal places (approximate). *(approximate)*

642. 703^2 (duplex method)

643. Estimate $\sqrt{20}$ to 2 decimal places (approximate). *(approximate)*

644. 383^2 (duplex method)

Answers: see the answer key (questions 637–644).

✎ Set 30: Squares, cubes and roots · Advanced · suggested time 16 min

645. Estimate √71 to 2 decimal places (approximate). *(approximate)*

646. 608^2 (duplex method)

647. Estimate √250 to 2 decimal places (approximate). *(approximate)*

648. 854^2 (duplex method)

649. Estimate √257 to 2 decimal places (approximate). *(approximate)*

650. 306^2 (duplex method)

651. Estimate √270 to 2 decimal places (approximate). *(approximate)*

652. 719^2 (duplex method)

Answers: see the answer key (questions 645–652).

Answers and explanations — Squares, cubes and roots

573. **2,025** — Digits before 5: 4 4 × (one more = 5) = 20 Attach 25 → 2,025

574. **1,681** — 41 × 41 = 1681.

575. **1,225** — Digits before 5: 3 3 × (one more = 4) = 12 Attach 25 → 1,225

576. **15,625** — Digits before 5: 12 12 × (one more = 13) = 156 Attach 25 → 15,625

577. **38,025** — Digits before 5: 19 19 × (one more = 20) = 380 Attach 25 → 38,025

578. **1,369** — 37 × 37 = 1369.

579. **5,625** — Digits before 5: 7 7 × (one more = 8) = 56 Attach 25 → 5,625

580. **24,025** — Digits before 5: 15 15 × (one more = 16) = 240 Attach 25 → 24,025

581. **361** — 19 × 19 = 361.

582. **784** — 28 × 28 = 784.

583. **11,025** — Digits before 5: 10 10 × (one more = 11) = 110 Attach 25 → 11,025

584. **256** — 16 × 16 = 256.

585. **30,625** — Digits before 5: 17 17 × (one more = 18) = 306 Attach 25 → 30,625

586. **289** — 17 × 17 = 289.

587. **9,025** — Digits before 5: 9 9 × (one more = 10) = 90 Attach 25 → 9,025

588. **900** — 30 × 30 = 900.

589. **34,225** — Digits before 5: 18 18 × (one more = 19) = 342 Attach 25 → 34,225

590. **1,521** — 39 × 39 = 1521.

591. **529** — 23 × 23 = 529.

592. **1,156** — 34 × 34 = 1156.

593. **3,364** — 58 × 58 = 3364.

594. **1,444** — 38 × 38 = 1444.

595. **2,601** — 51 × 51 = 2601.

596. **144** — 12 × 12 = 144.

597. **71** — Last digit 1 → root ends in 1 or 9. Leading part 50 lies between 7^2 and 8^2, so tens digit is 7. Check $75^2 = 5625$ to choose → 71.

598. **50** — Last digit 0 → cube root ends in 0 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 125 lies between 5^3 and 6^3 → tens digit 5.

599. **9,409** — $(100 - 3)^2 = 10000 - 2\times100\times3 + 9 = 9{,}409$.

600. **125** — 5 × 5 × 5 = 125.

601. **11,449** — Base 100, deviation +7 Left: 107 + 7 = 114 Right: $7^2 = 49$ (needs 2 digit(s)) Answer = 11,449

602. **41** — Last digit 1 → root ends in 1 or 9. Leading part 16 lies between 4^2 and 5^2, so tens digit is 4. Check $45^2 = 2025$ to choose → 41.

603. **33** — Last digit 7 → cube root ends in 3 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 35 lies between 3^3 and 4^3 → tens digit 3.

604. **3,969** — $(60 + 3)^2 = 3600 + 2\times60\times3 + 9 = 3{,}969$.

605. **90,601** — $(300 + 1)^2 = 90000 + 2\times300\times1 + 1 = 90{,}601$.

606. **12,996** — Base 100, deviation +14 Left: 114 + 14 = 128 Right: $14^2 = 196$ (needs 2 digit(s), carry 1) Answer = 12,996

607. **2,116** — Deviation from 50: −4 Left: 25 − 4 = 21 Right: $4^2 = 16$ Answer = 2,116

608. **47** — Last digit 9 → root ends in 3 or 7. Leading part 22 lies between 4^2 and 5^2, so tens digit is 4. Check $45^2 = 2025$ to choose → 47.

609. **66** — Last digit 6 → cube root ends in 6 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 287 lies between 6^3 and 7^3 → tens digit 6.

610. **4,761** — $(70 - 1)^2 = 4900 - 2\times70\times1 + 1 = 4{,}761$.

611. **13,824** — 24 × 24 × 24 = 13,824.

612. **9,21,600** — Base 1,000, deviation −40 Left: 960 − 40 = 920 Right: $40^2 = 1600$ (needs 3 digit(s), carry 1) Answer = 9,21,600

613. **6,724** — $(80 + 2)^2 = 6400 + 2\times80\times2 + 4 = 6{,}724$.

614. **9,56,484** — Base 1,000, deviation –22 Left: 978 – 22 = 956 Right: $22^2 = 484$ (needs 3 digit(s)) Answer = 9,56,484

615. **34** — Last digit 6 → root ends in 4 or 6. Leading part 11 lies between 3^2 and 4^2, so tens digit is 3. Check $35^2 = 1225$ to choose → 34.

616. **71** — Last digit 1 → cube root ends in 1 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 357 lies between 7^3 and 8^3 → tens digit 7.

617. **6,241** — $(80 - 1)^2 = 6400 - 2\times80\times1 + 1 = 6{,}241$.

618. **9,92,016** — Base 1,000, deviation –4 Left: 996 – 4 = 992 Right: $4^2 = 16$ (needs 3 digit(s)) Answer = 9,92,016

619. **59** — Last digit 1 → root ends in 1 or 9. Leading part 34 lies between 5^2 and 6^2, so tens digit is 5. Check $55^2 = 3025$ to choose → 59.

620. **23** — Last digit 7 → cube root ends in 3 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 12 lies between 2^3 and 3^3 → tens digit 2.

621. **8,649** — Base 100, deviation –7 Left: 93 – 7 = 86 Right: $7^2 = 49$ (needs 2 digit(s)) Answer = 8,649

622. **26** — Last digit 6 → root ends in 4 or 6. Leading part 6 lies between 2^2 and 3^2, so tens digit is 2. Check $25^2 = 625$ to choose → 26.

623. **48** — Last digit 2 → cube root ends in 8 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 110 lies between 4^3 and 5^3 → tens digit 4.

624. **91,204** — $(300 + 2)^2 = 90000 + 2\times300\times2 + 4 = 91{,}204$.

625. **70** — Last digit 0 → cube root ends in 0 (cube endings: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9). Thousands part 343 lies between 7^3 and 8^3 → tens digit 7.

626. **441** — $(20 + 1)^2 = 400 + 2\times20\times1 + 1 = 441$.

627. **1,000** — $10 \times 10 \times 10 = 1{,}000$.

628. **10,609** — Base 100, deviation +3 Left: 103 + 3 = 106 Right: $3^2 = 9$ (needs 2 digit(s)) Answer = 10,609

629. **≈ 16.91 (true value 16.912)** — Nearest perfect square 289 = 17^2. $\sqrt{n} \approx 17 + (286 - 289) \div (2 \times 17) = 16.912$.

630. **1,60,000** — Duplexes from left: 16, 0, 0, 0, 0 Combine with carries (right to left) → 1,60,000

631. **≈ 12.85 (true value 12.845)** — Nearest perfect square 169 = 13^2. $\sqrt{n} \approx 13 + (165 - 169) \div (2 \times 13) = 12.846$.

632. **9,29,296** — Duplexes from left: 81, 108, 108, 48, 16 Combine with carries (right to left) → 9,29,296

633. **≈ 19.39 (true value 19.391)** — Nearest perfect square 361 = 19^2. $\sqrt{n} \approx 19 + (376 - 361) \div (2 \times 19) = 19.395$.

634. **5,32,900** — Duplexes from left: 49, 42, 9, 0, 0 Combine with carries (right to left) → 5,32,900

635. **≈ 9.50 (true value 9.487)** — Nearest perfect square 81 = 9^2. $\sqrt{n} \approx 9 + (90 - 81) \div (2 \times 9) = 9.500$.

636. **11,881** — Duplexes from left: 1, 0, 18, 0, 81 Combine with carries (right to left) → 11,881

637. **2,83,024** — Duplexes from left: 25, 30, 29, 12, 4 Combine with carries (right to left) → 2,83,024

638. **≈ 8.78 (true value 8.775)** — Nearest perfect square 81 = 9^2. $\sqrt{n} \approx 9 + (77 - 81) \div (2 \times 9) = 8.778$.

639. **1,56,816** — Duplexes from left: 9, 54, 117, 108, 36 Combine with carries (right to left) → 1,56,816

640. **1,09,561** — Duplexes from left: 9, 18, 15, 6, 1 Combine with carries (right to left) → 1,09,561

641. **≈ 16.56 (true value 16.553)** — Nearest perfect square 289 = 17^2. $\sqrt{n} \approx 17 + (274 - 289) \div (2 \times 17) = 16.559$.

642. **4,94,209** — Duplexes from left: 49, 0, 42, 0, 9 Combine with carries (right to left) → 4,94,209

643. **≈ 4.50 (true value 4.472)** — Nearest perfect square 16 = 4^2. $\sqrt{n} \approx 4 + (20 - 16) \div (2 \times 4) = 4.500$.

644. **1,46,689** — Duplexes from left: 9, 48, 82, 48, 9 Combine with carries (right to left) → 1,46,689

645. **≈ 8.44 (true value 8.426)** — Nearest perfect square 64 = 8^2. $\sqrt{n} \approx 8 + (71 - 64) \div (2 \times 8) = 8.438$.

646. **3,69,664** — Duplexes from left: 36, 0, 96, 0, 64 Combine with carries (right to left) → 3,69,664

647. **≈ 15.81 (true value 15.811)** — Nearest perfect square 256 = 16^2. $\sqrt{n} \approx 16 + (250 - 256) \div (2 \times 16) = 15.812$.

648. **7,29,316** — Duplexes from left: 64, 80, 89, 40, 16 Combine with carries (right to left) → 7,29,316

649. **≈ 16.03 (true value 16.031)** — Nearest perfect square 256 = 16^2. $\sqrt{n} \approx 16 + (257 - 256) \div (2 \times 16) = 16.031$.

650. **93,636** — Duplexes from left: 9, 0, 36, 0, 36 Combine with carries (right to left) → 93,636

651. **≈ 16.44 (true value 16.432)** — Nearest perfect square 256 = 16^2. $\sqrt{n} \approx 16 + (270 - 256) \div (2 \times 16) = 16.438$.

652. **5,16,961** — Duplexes from left: 49, 14, 127, 18, 81 Combine with carries (right to left) → 5,16,961

Algebra and factorisation — 10 sets

✎ Set 31: Algebra and factorisation · Beginner · suggested time 6 min

653. Solve 3x + 18 = –3

654. Expand (x – 9)(3x + 2)

655. Solve 8x – 7 = 89

656. Expand (2x – 3)(3x + 9)

657. Solve 3x – 20 = –44

658. Expand (x + 1)(x – 9)

659. Solve 8x + 10 = 106

660. Expand (x + 6)(2x – 9)

Answers: see the answer key (questions 653–660).

✎ Set 32: Algebra and factorisation · Beginner · suggested time 6 min

661. Expand (4x – 2)(3x + 1)

662. Solve 8x – 13 = –29

663. Expand (2x + 5)(2x + 2)

664. Solve 7x – 7 = 56

665. Expand (3x + 9)(2x + 7)

666. Solve 2x – 15 = –39

667. Expand (x – 7)(x + 2)

668. Solve 8x + 14 = 94

Answers: see the answer key (questions 661–668).

✎ Set 33: Algebra and factorisation · Beginner · suggested time 6 min

669. Solve 7x + 1 = 99

670. Expand (2x + 8)(3x – 3)

671. Solve 9x + 3 = 111

672. Expand (3x – 1)(2x – 3)

673. Solve 8x – 11 = –83

674. Expand (x – 7)(3x + 7)

675. Solve 8x – 20 = 4

676. Expand (3x + 4)(x – 5)

Answers: see the answer key (questions 669–676).

✎ Set 34: Algebra and factorisation · Intermediate · suggested time 10 min

677. $80^2 - 16^2$

678. Use $(a \pm b)^2$ to find 41^2

679. Solve $12x - 7 = 6x + 29$

680. Solve: $5x + 8y = 61$, $2x + 1y = 9$

681. Factorise $x^2 - 8x + 7$

682. $49^2 - 6^2$

683. Use $(a \pm b)^2$ to find 58^2

684. Solve $8x - 7 = 2x - 19$

Answers: see the answer key (questions 677–684).

✎ Set 35: Algebra and factorisation · Intermediate · suggested time 10 min

685. Use $(a \pm b)^2$ to find 27^2

686. Solve $6x - 12 = 5x - 14$

687. Solve: $8x + 2y = 34$, $6x + 6y = 66$

688. Factorise $x^2 - 7x - 8$

689. $62^2 - 5^2$

690. Use $(a \pm b)^2$ to find 71^2

691. Solve $11x + 12 = 4x + 61$

692. Solve: $4x + 4y = -4$, $8x + 3y = 22$

Answers: see the answer key (questions 685–692).

✎ Set 36: Algebra and factorisation · Intermediate · suggested time 10 min

693. Solve $10x - 7 = 4x - 49$

694. Solve: $1x + 2y = 1$, $9x + 4y = 23$

695. Factorise $x^2 - 2x - 24$

696. $42^2 - 28^2$

697. Use $(a \pm b)^2$ to find 93^2

698. Solve $10x - 2 = 4x - 50$

699. Solve: $5x + 2y = 19$, $1x + 4y = 11$

700. Factorise $x^2 + 4x - 12$

Answers: see the answer key (questions 693–700).

✎ Set 37: Algebra and factorisation · Intermediate · suggested time 10 min

701. Solve: $7x + 3y = -51$, $5x + 6y = -48$

702. Factorise $x^2 + x - 6$

703. $50^2 - 26^2$

704. Use $(a \pm b)^2$ to find 68^2

705. Solve $8x + 13 = 4x + 21$

706. Solve: $1x + 4y = -19$, $7x + 4y = -37$

707. Factorise $x^2 - 13x + 36$

708. $86^2 - 23^2$

Answers: see the answer key (questions 701–708).

✎ Set 38: Algebra and factorisation · Advanced · suggested time 16 min

709. Solve $x^2 - 5x + 6 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

710. Solve: $15x + 48y = -93$, $48x + 15y = -159$

711. Solve $(x + 6)(x + 4) = (x + 3)(x + 12)$

712. Solve $x^3 + 15x^2 + 74x + 120 = 0$ by completing the cube

713. Solve $x^2 + 8x + 12 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

714. Solve: $47x + 46y = 135$, $46x + 47y = 144$

715. Solve $(x + 7)(x + 7) = (x + 1)(x + 1)$

716. Solve $x^2 + 0x - 1 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

Answers: see the answer key (questions 709–716).

✎ Set 39: Algebra and factorisation · Advanced · suggested time 16 min

717. Solve: $25x + 30y = 60$, $30x + 25y = 50$

718. Solve $(x + 1)(x + 2) = (x + 12)(x + 10)$

719. Solve $x^2 - 2x - 3 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

720. Solve: $50x + 59y = -436$, $59x + 50y = -436$

721. Solve $(x + 4)(x + 4) = (x + 11)(x + 6)$

722. Solve $x^2 - 10x + 21 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

723. Solve: $21x + 18y = 75$, $18x + 21y = 81$

724. Solve $(x + 2)(x + 6) = (x + 5)(x + 11)$

Answers: see the answer key (questions 717–724).

✎ Set 40: Algebra and factorisation - Advanced - suggested time 16 min

725. Solve $(x + 11)(x + 3) = (x + 9)(x + 9)$

726. Solve $x^2 - 1x - 56 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

727. Solve: $55x + 24y = -52$, $24x + 55y = 289$

728. Solve $(x + 10)(x + 9) = (x + 6)(x + 3)$

729. Solve $x^2 - 8x + 12 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

730. Solve: $13x + 26y = 130$, $26x + 13y = 221$

731. Solve $(x + 11)(x + 10) = (x + 1)(x + 12)$

732. Solve $x^2 + 8x + 15 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

Answers: see the answer key (questions 725–732).

Answers and explanations — Algebra and factorisation

653. **x = −7** — Transpose 18 (change sign): 3x = −3 − 18 = −21; divide by 3: x = −7.

654. **3x² − 25x − 18** — Vertical (left): 1 × 3 = 3. Crosswise: 1 × (2) + 3 × (−9) = −25. Vertical (right): (−9) × (2) = −18.

655. **x = 12** — Transpose −7 (change sign): 8x = 89 + 7 = 96; divide by 8: x = 12.

656. **6x² + 9x − 27** — Vertical (left): 2 × 3 = 6. Crosswise: 2 × (9) + 3 × (−3) = 9. Vertical (right): (−3) × (9) = −27.

657. **x = −8** — Transpose −20 (change sign): 3x = −44 + 20 = −24; divide by 3: x = −8.

658. **x² − 8x − 9** — Vertical (left): 1 × 1 = 1. Crosswise: 1 × (−9) + 1 × (1) = −8. Vertical (right): (1) × (−9) = −9.

659. **x = 12** — Transpose 10 (change sign): 8x = 106 − 10 = 96; divide by 8: x = 12.

660. **2x² + 3x − 54** — Vertical (left): 1 × 2 = 2. Crosswise: 1 × (−9) + 2 × (6) = 3. Vertical (right): (6) × (−9) = −54.

661. **12x² − 2x − 2** — Vertical (left): 4 × 3 = 12. Crosswise: 4 × (1) + 3 × (−2) = −2. Vertical (right): (−2) × (1) = −2.

662. **x = −2** — Transpose −13 (change sign): 8x = −29 + 13 = −16; divide by 8: x = −2.

663. **4x² + 14x + 10** — Vertical (left): 2 × 2 = 4. Crosswise: 2 × (2) + 2 × (5) = 14. Vertical (right): (5) × (2) = 10.

664. **x = 9** — Transpose −7 (change sign): 7x = 56 + 7 = 63; divide by 7: x = 9.

665. **6x² + 39x + 63** — Vertical (left): 3 × 2 = 6. Crosswise: 3 × (7) + 2 × (9) = 39. Vertical (right): (9) × (7) = 63.

666. **x = −12** — Transpose −15 (change sign): 2x = −39 + 15 = −24; divide by 2: x = −12.

667. **x² − 5x − 14** — Vertical (left): 1 × 1 = 1. Crosswise: 1 × (2) + 1 × (−7) = −5. Vertical (right): (−7) × (2) = −14.

668. **x = 10** — Transpose 14 (change sign): 8x = 94 − 14 = 80; divide by 8: x = 10.

669. **x = 14** — Transpose 1 (change sign): 7x = 99 − 1 = 98; divide by 7: x = 14.

670. **6x² + 18x − 24** — Vertical (left): 2 × 3 = 6. Crosswise: 2 × (−3) + 3 × (8) = 18. Vertical (right): (8) × (−3) = −24.

671. **x = 12** — Transpose 3 (change sign): 9x = 111 − 3 = 108; divide by 9: x = 12.

672. **6x² − 11x + 3** — Vertical (left): 3 × 2 = 6. Crosswise: 3 × (−3) + 2 × (−1) = −11. Vertical (right): (−1) × (−3) = 3.

673. **x = −9** — Transpose −11 (change sign): 8x = −83 + 11 = −72; divide by 8: x = −9.

674. **3x² − 14x − 49** — Vertical (left): 1 × 3 = 3. Crosswise: 1 × (7) + 3 × (−7) = −14. Vertical (right): (−7) × (7) = −49.

675. **x = 3** — Transpose −20 (change sign): 8x = 4 + 20 = 24; divide by 8: x = 3.

676. **3x² − 11x − 20** — Vertical (left): 3 × 1 = 3. Crosswise: 3 × (−5) + 1 × (4) = −11. Vertical (right): (4) × (−5) = −20.

677. **6,144** — $a^2 - b^2 = (a + b)(a - b) = 96 \times 64 = 6{,}144$.

678. **1,681** — $(40 + 1)^2 = 1600 + 2{\times}40{\times}1 + 1 = 1{,}681$.

679. **x = 6** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (29 − −7)/(12 − 6) = 36/6 = 6.

680. **x = 1, y = 7** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (61 - 72)/-11 = 1$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (45 - 122)/-11 = 7$.

681. **(x − 1)(x − 7)** — Find two numbers with product 7 and sum −8: −1 and −7. Check with Gunita-samuccaya (put x = 1): (1 − 1)(1 − 7) = 0 and 1 − 8 + 7 = 0.

682. **2,365** — $a^2 - b^2 = (a + b)(a - b) = 55 \times 43 = 2{,}365$.

683. **3,364** — $(60 - 2)^2 = 3600 - 2{\times}60{\times}2 + 4 = 3{,}364$.

684. **x = −2** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−19 − −7)/(8 − 2) = −12/6 = −2.

685. **729** — $(30 - 3)^2 = 900 - 2{\times}30{\times}3 + 9 = 729$.

686. **x = −2** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−14 − −12)/(6 − 5) = −2/1 = −2.

687. **x = 2, y = 9** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (204 - 132)/36 = 2$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (528 - 204)/36 = 9$.

688. **(x + 1)(x − 8)** — Find two numbers with product −8 and sum −7: 1 and −8. Check with Gunita-samuccaya (put x = 1): (1 + 1)(1 − 8) = −14 and 1 − 7 − 8 = −14.

689. **3,819** — $a^2 - b^2 = (a + b)(a - b) = 67 \times 57 = 3{,}819$.

690. **5,041** — $(70 + 1)^2 = 4900 + 2{\times}70{\times}1 + 1 = 5{,}041$.

691. **x = 7** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (61 − 12)/(11 − 4) = 49/7 = 7.

692. **x = 5, y = −6** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (-12 - 88)/-20 = 5$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (88 - -32)/-20 = -6$.

693. **x = −7** — Paravartya formula for ax + b = cx + d: x = (d − b)/(a − c) = (−49 − −7)/(10 − 4) = −42/6 = −7.

694. **x = 3, y = −1** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (4 - 46)/-14 = 3$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (23 - 9)/-14 = -1$.

695. **(x + 4)(x – 6)** — Find two numbers with product –24 and sum –2: 4 and –6. Check with Gunita-samuccaya (put x = 1): (1 + 4)(1 – 6) = –25 and 1 – 2 – 24 = –25.

696. **980** — $a^2 - b^2 = (a + b)(a - b) = 70 \times 14 = 980$.

697. **8,649** — $(90 + 3)^2 = 8100 + 2{\times}90{\times}3 + 9 = 8{,}649$.

698. **x = –8** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (–50 – –2)/(10 – 4) = –48/6 = –8.

699. **x = 3, y = 2** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (76 - 22)/18 = 3$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (55 - 19)/18 = 2$.

700. **(x – 2)(x + 6)** — Find two numbers with product –12 and sum 4: –2 and 6. Check with Gunita-samuccaya (put x = 1): (1 – 2)(1 + 6) = –7 and 1 + 4 – 12 = –7.

701. **x = –6, y = –3** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (–306 - –144)/27 = –6$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (–336 - –255)/27 = –3$.

702. **(x – 2)(x + 3)** — Find two numbers with product –6 and sum 1: –2 and 3. Check with Gunita-samuccaya (put x = 1): (1 – 2)(1 + 3) = –4 and 1 + 1 – 6 = –4.

703. **1,824** — $a^2 - b^2 = (a + b)(a - b) = 76 \times 24 = 1{,}824$.

704. **4,624** — $(70 - 2)^2 = 4900 - 2{\times}70{\times}2 + 4 = 4{,}624$.

705. **x = 2** — Paravartya formula for ax + b = cx + d: x = (d – b)/(a – c) = (21 – 13)/(8 – 4) = 8/4 = 2.

706. **x = –3, y = –4** — $x = (c_1b_2 - c_2b_1)/(a_1b_2 - a_2b_1) = (–76 - –148)/–24 = –3$; $y = (a_1c_2 - a_2c_1)/(a_1b_2 - a_2b_1) = (–37 - –133)/–24 = –4$.

707. **(x – 4)(x – 9)** — Find two numbers with product 36 and sum –13: –4 and –9. Check with Gunita-samuccaya (put x = 1): (1 – 4)(1 – 9) = 24 and 1 – 13 + 36 = 24.

708. **6,867** — $a^2 - b^2 = (a + b)(a - b) = 109 \times 63 = 6{,}867$.

709. **x = 3 or 2** — Differential: 2x – 5. Discriminant = 25 – 24 = 1, √ = 1. 2x – 5 = ±1.

710. **x = –3, y = –1** — Add: 63(x + y) = –252 → x + y = –4. Subtract: -33(x – y) = 66 → x – y = –2. Solve: x = –3, y = –1.

711. **x = –2 2/5** — x = (cd – ab)/(a + b – c – d) = (36 – 24)/(10 – 15).

712. **x = –6, –5, –4** — $(x + 5)^3 = x^3 + 15x^2 + 75x + 125$. The given cubic = $(x + 5)^3 - (x + 5)$. Let y = x + 5: $y^3 - y = 0 \rightarrow y = 0, \pm 1$.

713. **x = –2 or –6** — Differential: 2x + 8. Discriminant = 64 – 48 = 16, √ = 4. 2x + 8 = ±4.

714. **x = –3, y = 6** — Add: 93(x + y) = 279 → x + y = 3. Subtract: 1(x – y) = –9 → x – y = –9. Solve: x = –3, y = 6.

715. **x = –4** — x = (cd – ab)/(a + b – c – d) = (1 – 49)/(14 – 2).

716. **x = 1 or –1** — Differential: 2x + 0. Discriminant = 0 – -4 = 4, √ = 2. 2x + 0 = ±2.

717. **x = 0, y = 2** — Add: 55(x + y) = 110 → x + y = 2. Subtract: -5(x – y) = 10 → x – y = –2. Solve: x = 0, y = 2.

718. **x = –6 4/19** — x = (cd – ab)/(a + b – c – d) = (120 – 2)/(3 – 22).

719. **x = 3 or –1** — Differential: 2x – 2. Discriminant = 4 – -12 = 16, √ = 4. 2x – 2 = ±4.

720. **x = –4, y = –4** — Add: 109(x + y) = –872 → x + y = –8. Subtract: -9(x – y) = 0 → x – y = 0. Solve: x = –4, y = –4.

721. **x = –5 5/9** — x = (cd – ab)/(a + b – c – d) = (66 – 16)/(8 – 17).

722. **x = 7 or 3** — Differential: 2x – 10. Discriminant = 100 – 84 = 16, √ = 4. 2x – 10 = ±4.

723. **x = 1, y = 3** — Add: 39(x + y) = 156 → x + y = 4. Subtract: 3(x – y) = –6 → x – y = –2. Solve: x = 1, y = 3.

724. **x = –5 3/8** — x = (cd – ab)/(a + b – c – d) = (55 – 12)/(8 – 16).

725. **x = –12** — x = (cd – ab)/(a + b – c – d) = (81 – 33)/(14 – 18).

726. **x = 8 or –7** — Differential: 2x – 1. Discriminant = 1 – -224 = 225, √ = 15. 2x – 1 = ±15.

727. **x = –4, y = 7** — Add: 79(x + y) = 237 → x + y = 3. Subtract: 31(x – y) = –341 → x – y = –11. Solve: x = –4, y = 7.

728. **x = –7 1/5** — x = (cd – ab)/(a + b – c – d) = (18 – 90)/(19 – 9).

729. **x = 6 or 2** — Differential: 2x – 8. Discriminant = 64 – 48 = 16, √ = 4. 2x – 8 = ±4.

730. **x = 8, y = 1** — Add: 39(x + y) = 351 → x + y = 9. Subtract: -13(x – y) = –91 → x – y = 7. Solve: x = 8, y = 1.

731. **x = –12 1/4** — x = (cd – ab)/(a + b – c – d) = (12 – 110)/(21 – 13).

732. **x = –3 or –5** — Differential: 2x + 8. Discriminant = 64 – 60 = 4, √ = 2. 2x + 8 = ±2.

Percentages, ratios and averages — 10 sets

✎ Set 41: Percentages, ratios and averages · Beginner · suggested time 6 min

733. Find 20% of 800.

734. Write 5/6 as a percentage.

735. Find 25% of 124 using a fraction.

736. Divide ₹11,835 between A and B in the ratio 8 : 7.

737. Find the average of 62, 65, 63, 65, 60.

738. Find 120% of 80.

739. Write 1/9 as a percentage.

740. Find 33 1/3% of 186 using a fraction.

Answers: see the answer key (questions 733–740).

✎ Set 42: Percentages, ratios and averages · Beginner · suggested time 6 min

741. Write 5/8 as a percentage.

742. Find 37 1/2% of 656 using a fraction.

743. Divide ₹2,259 between A and B in the ratio 4 : 5.

744. Find the average of 48, 57, 45, 46, 58, 52.

745. Find 120% of 480.

746. Write 2/3 as a percentage.

747. Find 11 1/9% of 774 using a fraction.

748. Divide ₹884 between A and B in the ratio 3 : 1.

Answers: see the answer key (questions 741–748).

✎ Set 43: Percentages, ratios and averages · Beginner · suggested time 6 min

749. Find 75% of 488 using a fraction.

750. Divide ₹2,540 between A and B in the ratio 2 : 3.

751. Find the average of 80, 68, 80, 68, 59.

752. Find 120% of 7,200.

753. Write 1/3 as a percentage.

754. Find 33 1/3% of 174 using a fraction.

755. Divide ₹7,150 between A and B in the ratio 7 : 3.

756. Find the average of 57, 52, 49, 64, 53.

Answers: see the answer key (questions 749–756).

✎ Set 44: Percentages, ratios and averages · Intermediate · suggested time 10 min

757. Find the single discount equal to successive discounts of 10% and 15%.

758. A invests ₹20,000 for 12 months and B invests ₹25,000 for 10 months. Profit is ₹61,201. Find A's share.

759. A price changes from ₹1,600 to ₹1,760. Find the percentage increase.

760. Section A has 25 students with average 54; section B has 28 students with average 69. Find the combined average.

761. An item costing ₹200 is sold for ₹190. Find the loss percentage.

762. Find the single discount equal to successive discounts of 30% and 10%.

763. A invests ₹30,000 for 12 months and B invests ₹35,000 for 10 months. Profit is ₹1,13,387. Find A's share.

764. A price changes from ₹1,600 to ₹1,360. Find the percentage decrease.

Answers: see the answer key (questions 757–764).

✎ Set 45: Percentages, ratios and averages - Intermediate - suggested time 10 min

765. A invests ₹20,000 for 12 months and B invests ₹60,000 for 10 months. Profit is ₹14,210. Find A's share.

766. A price changes from ₹4,000 to ₹5,000. Find the percentage increase.

767. Section A has 33 students with average 58; section B has 38 students with average 65. Find the combined average.

768. An item costing ₹1,500 is sold for ₹1,680. Find the profit percentage.

769. Find the single discount equal to successive discounts of 10% and 10%.

770. A invests ₹50,000 for 12 months and B invests ₹60,000 for 12 months. Profit is ₹11,154. Find A's share.

771. A price changes from ₹400 to ₹368. Find the percentage decrease.

772. Section A has 29 students with average 67; section B has 40 students with average 69. Find the combined average.

Answers: see the answer key (questions 765–772).

✎ Set 46: Percentages, ratios and averages - Intermediate - suggested time 10 min

773. A price changes from ₹1,250 to ₹750. Find the percentage decrease.

774. Section A has 37 students with average 60; section B has 37 students with average 65. Find the combined average.

775. An item costing ₹1,200 is sold for ₹1,260. Find the profit percentage.

776. Find the single discount equal to successive discounts of 25% and 20%.

777. A invests ₹20,000 for 12 months and B invests ₹25,000 for 9 months. Profit is ₹25,451. Find A's share.

778. A price changes from ₹200 to ₹250. Find the percentage increase.

779. Section A has 10 students with average 62; section B has 20 students with average 55. Find the combined average.

780. An item costing ₹800 is sold for ₹600. Find the loss percentage.

Answers: see the answer key (questions 773–780).

✎ Set 47: Percentages, ratios and averages · Intermediate · suggested time 10 min

781. Section A has 19 students with average 61; section B has 35 students with average 62. Find the combined average.

782. An item costing ₹800 is sold for ₹960. Find the profit percentage.

783. Find the single discount equal to successive discounts of 30% and 15%.

784. A invests ₹20,000 for 12 months and B invests ₹60,000 for 6 months. Profit is ₹10,890. Find A's share.

785. A price changes from ₹1,250 to ₹1,125. Find the percentage decrease.

786. Section A has 19 students with average 58; section B has 30 students with average 55. Find the combined average.

787. An item costing ₹200 is sold for ₹240. Find the profit percentage.

788. Find the single discount equal to successive discounts of 25% and 10%.

Answers: see the answer key (questions 781–788).

✎ Set 48: Percentages, ratios and averages - Advanced - suggested time 16 min

789. In what ratio should rice at ₹30/kg be mixed with rice at ₹90/kg to get a mixture worth ₹49/kg?

790. In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹58/kg?

791. In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹49/kg?

792. In what ratio should rice at ₹20/kg be mixed with rice at ₹70/kg to get a mixture worth ₹61/kg?

793. In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹44/kg?

794. In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹65/kg?

795. In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹62/kg?

796. In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹43/kg?

Answers: see the answer key (questions 789–796).

✎ Set 49: Percentages, ratios and averages - Advanced - suggested time 16 min

797. In what ratio should rice at ₹50/kg be mixed with rice at ₹80/kg to get a mixture worth ₹54/kg?

798. In what ratio should rice at ₹20/kg be mixed with rice at ₹80/kg to get a mixture worth ₹39/kg?

799. In what ratio should rice at ₹50/kg be mixed with rice at ₹90/kg to get a mixture worth ₹76/kg?

800. In what ratio should rice at ₹30/kg be mixed with rice at ₹90/kg to get a mixture worth ₹86/kg?

801. In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹39/kg?

802. In what ratio should rice at ₹20/kg be mixed with rice at ₹70/kg to get a mixture worth ₹38/kg?

803. In what ratio should rice at ₹20/kg be mixed with rice at ₹90/kg to get a mixture worth ₹34/kg?

804. In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹58/kg?

Answers: see the answer key (questions 797–804).

✎ Set 50: Percentages, ratios and averages - Advanced - suggested time 16 min

805. In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹50/kg?

806. In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹53/kg?

807. In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹45/kg?

808. In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹48/kg?

809. In what ratio should rice at ₹20/kg be mixed with rice at ₹70/kg to get a mixture worth ₹29/kg?

810. In what ratio should rice at ₹30/kg be mixed with rice at ₹90/kg to get a mixture worth ₹54/kg?

811. In what ratio should rice at ₹30/kg be mixed with rice at ₹80/kg to get a mixture worth ₹70/kg?

812. In what ratio should rice at ₹50/kg be mixed with rice at ₹80/kg to get a mixture worth ₹73/kg?

Answers: see the answer key (questions 805–812).

Answers and explanations — Percentages, ratios and averages

733. **160** — Split 20% into easy parts (10%, 5%, 1% …) or use fractions: 20% = 1/5. Result 160.

734. **83 1/3%** — 5/6 × 100 = 83 1/3%.

735. **31** — 25% = 1/4. 1/4 × 124 = 31.

736. **A ₹6,312, B ₹5,523** — Total parts 15; one part = 11,835 ÷ 15 = 789.

737. **63** — Assume 63. Deviations: −1, +2, +0, +2, −3; total +0; ÷ 5 = 0. Average = 63.

738. **96** — Split 120% into easy parts (10%, 5%, 1% …) or use fractions: 120% = 6/5. Result 96.

739. **11 1/9%** — 1/9 × 100 = 11 1/9%.

740. **62** — 33 1/3% = 1/3. 1/3 × 186 = 62.

741. **62 1/2%** — 5/8 × 100 = 62 1/2%.

742. **246** — 37 1/2% = 3/8. 3/8 × 656 = 246.

743. **A ₹1,004, B ₹1,255** — Total parts 9; one part = 2,259 ÷ 9 = 251.

744. **51** — Assume 52. Deviations: −4, +5, −7, −6, +6, +0; total −6; ÷ 6 = −1. Average = 51.

745. **576** — Split 120% into easy parts (10%, 5%, 1% …) or use fractions: 120% = 6/5. Result 576.

746. **66 2/3%** — 2/3 × 100 = 66 2/3%.

747. **86** — 11 1/9% = 1/9. 1/9 × 774 = 86.

748. **A ₹663, B ₹221** — Total parts 4; one part = 884 ÷ 4 = 221.

749. **366** — 75% = 3/4. 3/4 × 488 = 366.

750. **A ₹1,016, B ₹1,524** — Total parts 5; one part = 2,540 ÷ 5 = 508.

751. **71** — Assume 71. Deviations: +9, −3, +9, −3, −12; total +0; ÷ 5 = 0. Average = 71.

752. **8,640** — Split 120% into easy parts (10%, 5%, 1% …) or use fractions: 120% = 6/5. Result 8,640.

753. **33 1/3%** — 1/3 × 100 = 33 1/3%.

754. **58** — 33 1/3% = 1/3. 1/3 × 174 = 58.

755. **A ₹5,005, B ₹2,145** — Total parts 10; one part = 7,150 ÷ 10 = 715.

756. **55** — Assume 57. Deviations: +0, −5, −8, +7, −4; total −10; ÷ 5 = −2. Average = 55.

757. **23 1/2%** — Net = a + b − ab/100 = 10 + 15 − 150/100 = 23 1/2%.

758. **₹29,976** — Ratio = 20,000×12 : 25,000×10 = 24 : 25. A's share = 24/49 × 61,201 = 29,976.

759. **10%** — Change = 160. % change = change ÷ original × 100 = 160/1600 × 100 = 10%.

760. **61.92** — (25×54 + 28×69) ÷ 53 = 3282 ÷ 53.

761. **5% loss** — Loss = 10. % = 10/200 × 100 = 5%.

762. **37%** — Net = a + b − ab/100 = 30 + 10 − 300/100 = 37%.

763. **₹57,492** — Ratio = 30,000×12 : 35,000×10 = 36 : 35. A's share = 36/71 × 1,13,387 = 57,492.

764. **15%** — Change = 240. % change = change ÷ original × 100 = 240/1600 × 100 = 15%.

765. **₹4,060** — Ratio = 20,000×12 : 60,000×10 = 2 : 5. A's share = 2/7 × 14,210 = 4,060.

766. **25%** — Change = 1,000. % change = change ÷ original × 100 = 1000/4000 × 100 = 25%.

767. **61.75** — (33×58 + 38×65) ÷ 71 = 4384 ÷ 71.

768. **12% profit** — Profit = 180. % = 180/1500 × 100 = 12%.

769. **19%** — Net = a + b − ab/100 = 10 + 10 − 100/100 = 19%.

770. **₹5,070** — Ratio = 50,000×12 : 60,000×12 = 5 : 6. A's share = 5/11 × 11,154 = 5,070.

771. **8%** — Change = 32. % change = change ÷ original × 100 = 32/400 × 100 = 8%.

772. **68.16** — (29×67 + 40×69) ÷ 69 = 4703 ÷ 69.

773. **40%** — Change = 500. % change = change ÷ original × 100 = 500/1250 × 100 = 40%.

774. **62.50** — (37×60 + 37×65) ÷ 74 = 4625 ÷ 74.

775. **5% profit** — Profit = 60. % = 60/1200 × 100 = 5%.

776. **40%** — Net = a + b − ab/100 = 25 + 20 − 500/100 = 40%.

777. **₹13,136** — Ratio = 20,000×12 : 25,000×9 = 16 : 15. A's share = 16/31 × 25,451 = 13,136.

778. **25%** — Change = 50. % change = change ÷ original × 100 = 50/200 × 100 = 25%.

779. **57.33** — (10×62 + 20×55) ÷ 30 = 1720 ÷ 30.

780. **25% loss** — Loss = 200. % = 200/800 × 100 = 25%.

781. **61.65** — (19×61 + 35×62) ÷ 54 = 3329 ÷ 54.

782. **20% profit** — Profit = 160. % = 160/800 × 100 = 20%.

783. **40 1/2%** — Net = a + b − ab/100 = 30 + 15 − 450/100 = 40 1/2%.

784. **₹4,356** — Ratio = 20,000×12 : 60,000×6 = 2 : 3. A's share = 2/5 × 10,890 = 4,356.

785. **10%** — Change = 125. % change = change ÷ original × 100 = 125/1250 × 100 = 10%.

786. **56.16** — (19×58 + 30×55) ÷ 49 = 2752 ÷ 49.

787. **20% profit** — Profit = 40. % = 40/200 × 100 = 20%.

788. **32 1/2%** — Net = a + b − ab/100 = 25 + 10 − 250/100 = 32 1/2%.

789. **41 : 19** — Alligation cross: (90 − 49) : (49 − 30) = 41 : 19 = 41 : 19.

790. **3 : 2** — Alligation cross: (70 − 58) : (58 − 50) = 12 : 8 = 3 : 2.

791. **11 : 9** — Alligation cross: (60 − 49) : (49 − 40) = 11 : 9 = 11 : 9.

792. **9 : 41** — Alligation cross: (70 − 61) : (61 − 20) = 9 : 41 = 9 : 41.

793. **4 : 1** — Alligation cross: (60 − 44) : (44 − 40) = 16 : 4 = 4 : 1.

794. **1 : 3** — Alligation cross: (70 − 65) : (65 − 50) = 5 : 15 = 1 : 3.

795. **2 : 3** — Alligation cross: (70 − 62) : (62 − 50) = 8 : 12 = 2 : 3.

796. **9 : 1** — Alligation cross: (70 − 43) : (43 − 40) = 27 : 3 = 9 : 1.

797. **13 : 2** — Alligation cross: (80 − 54) : (54 − 50) = 26 : 4 = 13 : 2.

798. **41 : 19** — Alligation cross: (80 − 39) : (39 − 20) = 41 : 19 = 41 : 19.

799. **7 : 13** — Alligation cross: (90 − 76) : (76 − 50) = 14 : 26 = 7 : 13.

800. **1 : 14** — Alligation cross: (90 − 86) : (86 − 30) = 4 : 56 = 1 : 14.

801. **7 : 3** — Alligation cross: (60 − 39) : (39 − 30) = 21 : 9 = 7 : 3.

802. **16 : 9** — Alligation cross: (70 − 38) : (38 − 20) = 32 : 18 = 16 : 9.

803. **4 : 1** — Alligation cross: (90 − 34) : (34 − 20) = 56 : 14 = 4 : 1.

804. **2 : 3** — Alligation cross: (70 − 58) : (58 − 40) = 12 : 18 = 2 : 3.

805. **1 : 2** — Alligation cross: (60 − 50) : (50 − 30) = 10 : 20 = 1 : 2.

806. **27 : 23** — Alligation cross: (80 − 53) : (53 − 30) = 27 : 23 = 27 : 23.

807. **5 : 1** — Alligation cross: (70 − 45) : (45 − 40) = 25 : 5 = 5 : 1.

808. **21 : 4** — Alligation cross: (90 − 48) : (48 − 40) = 42 : 8 = 21 : 4.

809. **41 : 9** — Alligation cross: (70 − 29) : (29 − 20) = 41 : 9 = 41 : 9.

810. **3 : 2** — Alligation cross: (90 − 54) : (54 − 30) = 36 : 24 = 3 : 2.

811. **1 : 4** — Alligation cross: (80 − 70) : (70 − 30) = 10 : 40 = 1 : 4.

812. **7 : 23** — Alligation cross: (80 − 73) : (73 − 50) = 7 : 23 = 7 : 23.

Government-exam mixed aptitude — 10 sets

✎ Set 51: Government-exam mixed aptitude · Beginner · suggested time 6 min

813. Find the simple interest on ₹25,000 at 5% per annum for 4 years.

814. A car travels at 72 km/h for 2 hours. Find the distance, and the speed in m/s.

815. Divide ₹4,356 between A and B in the ratio 2 : 9.

816. Find the simple interest on ₹5,000 at 8% per annum for 5 years.

817. A car travels at 45 km/h for 2 hours. Find the distance, and the speed in m/s.

818. Divide ₹2,688 between A and B in the ratio 5 : 3.

819. Find the simple interest on ₹15,000 at 6% per annum for 2 years.

820. A car travels at 90 km/h for 1 1/2 hours. Find the distance, and the speed in m/s.

Answers: see the answer key (questions 813–820).

✎ Set 52: Government-exam mixed aptitude · Beginner · suggested time 6 min

821. Divide ₹8,382 between A and B in the ratio 7 : 4.

822. Find the simple interest on ₹8,000 at 10% per annum for 2 years.

823. A car travels at 36 km/h for 2 hours. Find the distance, and the speed in m/s.

824. Divide ₹11,115 between A and B in the ratio 4 : 9.

825. Find the simple interest on ₹12,000 at 5% per annum for 2 years.

826. Divide ₹7,304 between A and B in the ratio 4 : 7.

827. Find the simple interest on ₹40,000 at 8% per annum for 6 years.

828. Divide ₹7,592 between A and B in the ratio 8 : 5.

Answers: see the answer key (questions 821–828).

✎ Set 53: Government-exam mixed aptitude · Beginner · suggested time 6 min

829. Divide ₹4,125 between A and B in the ratio 3 : 8.

830. Find the simple interest on ₹40,000 at 10% per annum for 6 years.

831. A car travels at 90 km/h for 3/4 hours. Find the distance, and the speed in m/s.

832. Divide ₹3,563 between A and B in the ratio 4 : 3.

833. Find the simple interest on ₹25,000 at 6% per annum for 6 years.

834. A car travels at 60 km/h for 2 1/2 hours. Find the distance, and the speed in m/s.

835. Divide ₹1,908 between A and B in the ratio 7 : 2.

836. Find the simple interest on ₹5,000 at 10% per annum for 2 years.

Answers: see the answer key (questions 829–836).

✎ Set 54: Government-exam mixed aptitude · Intermediate · suggested time 10 min

837. Find the compound interest on ₹80,000 at 20% per annum, compounded yearly, for 2 years.

838. A can finish a job in 20 days and B in 5 days. How long will they take together?

839. Two trains 200 m and 80 m long run at 72 km/h and 45 km/h in opposite directions. How long do they take to cross each other?

840. A boat goes downstream at 21 km/h and upstream at 13 km/h. Find the speed of the boat in still water and of the stream.

841. A father is 2 times as old as his son. In 2 years the sum of their ages will be 37. Find their present ages.

842. Find the area and circumference of a circle of radius 28 cm ($\pi = 22/7$).

843. A card is drawn from a well-shuffled deck of 52. Probability of a king?

844. In how many ways can 2 prizes (1st, 2nd, …) be given to 8 students, one prize each?

Answers: see the answer key (questions 837–844).

✎ Set 55: Government-exam mixed aptitude · Intermediate · suggested time 10 min

845. Pipe A fills a tank in 5 hours; pipe B empties it in 20 hours. If both are open, how long to fill the empty tank?

846. Two trains 150 m and 130 m long run at 54 km/h and 36 km/h in opposite directions. How long do they take to cross each other?

847. A boat goes downstream at 16 km/h and upstream at 4 km/h. Find the speed of the boat in still water and of the stream.

848. A father is 4 times as old as his son. In 10 years the sum of their ages will be 70. Find their present ages.

849. Volume of a cone with radius 21 cm and height 6 cm ($\pi = 22/7$)?

850. 2 fair coins are tossed. Probability of exactly 2 heads?

851. In how many ways can a committee of 2 be chosen from 7 people?

852. Find the last digit of 13^{724}.

Answers: see the answer key (questions 845–852).

✎ Set 56: Government-exam mixed aptitude · Intermediate · suggested time 10 min

853. Pipe A fills a tank in 8 hours; pipe B empties it in 24 hours. If both are open, how long to fill the empty tank?

854. Two trains 100 m and 130 m long run at 90 km/h and 45 km/h in opposite directions. How long do they take to cross each other?

855. A boat goes downstream at 15 km/h and upstream at 11 km/h. Find the speed of the boat in still water and of the stream.

856. A father is 2 times as old as his son. In 5 years the sum of their ages will be 67. Find their present ages.

857. Volume of a cylinder with radius 7 cm and height 9 cm ($\pi = 22/7$)?

858. A card is drawn from a well-shuffled deck of 52. Probability of a red card?

859. In how many ways can a committee of 4 be chosen from 12 people?

860. Find the last digit of 8^{799}.

Answers: see the answer key (questions 853–860).

✎ Set 57: Government-exam mixed aptitude · Intermediate · suggested time 10 min

861. Two trains 240 m and 130 m long run at 72 km/h and 45 km/h in opposite directions. How long do they take to cross each other?

862. A boat goes downstream at 18 km/h and upstream at 16 km/h. Find the speed of the boat in still water and of the stream.

863. A father is 4 times as old as his son. In 4 years the sum of their ages will be 108. Find their present ages.

864. A rectangular field is 29 m by 14 m. Find its area and perimeter.

865. In how many ways can 4 prizes (1st, 2nd, …) be given to 12 students, one prize each?

866. Find the last digit of 22^{22}.

867. Find the remainder when 97,832 is divided by 9.

868. Simplify: $12 + 6 \div 3 \times 2 - 13$

Answers: see the answer key (questions 861–868).

✎ Set 58: Government-exam mixed aptitude · Advanced · suggested time 16 min

869. In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹56/kg?

870. Approximate: 887.97 × 33.02 ÷ 10 + 138.1 *(approximate)*

871. In what ratio should rice at ₹20/kg be mixed with rice at ₹80/kg to get a mixture worth ₹36/kg?

872. Approximate: 711.03 × 24.02 ÷ 10 + 677.1 *(approximate)*

873. In what ratio should rice at ₹30/kg be mixed with rice at ₹60/kg to get a mixture worth ₹56/kg?

874. Approximate: 337.03 × 23.98 ÷ 10 + 397.1 *(approximate)*

875. In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹48/kg?

876. Approximate: 356.03 × 47.98 ÷ 10 + 321.1 *(approximate)*

Answers: see the answer key (questions 869–876).

✎ Set 59: Government-exam mixed aptitude · Advanced · suggested time 16 min

877. Approximate: 412.98 × 46.98 ÷ 10 + 833.9 *(approximate)*

878. In what ratio should rice at ₹50/kg be mixed with rice at ₹70/kg to get a mixture worth ₹59/kg?

879. Approximate: 286.02 × 26.02 ÷ 10 + 461.9 *(approximate)*

880. In what ratio should rice at ₹40/kg be mixed with rice at ₹90/kg to get a mixture worth ₹67/kg?

881. Approximate: 396.02 × 30.02 ÷ 10 + 434.9 *(approximate)*

882. In what ratio should rice at ₹30/kg be mixed with rice at ₹90/kg to get a mixture worth ₹67/kg?

883. Approximate: 579.03 × 36.02 ÷ 10 + 569.1 *(approximate)*

884. In what ratio should rice at ₹30/kg be mixed with rice at ₹70/kg to get a mixture worth ₹37/kg?

Answers: see the answer key (questions 877–884).

✎ Set 60: Government-exam mixed aptitude - Advanced - suggested time 16 min

885. In what ratio should rice at ₹20/kg be mixed with rice at ₹80/kg to get a mixture worth ₹27/kg?

886. Approximate: 891.98 × 21.98 ÷ 10 + 418.9 *(approximate)*

887. In what ratio should rice at ₹40/kg be mixed with rice at ₹70/kg to get a mixture worth ₹51/kg?

888. Approximate: 692.03 × 23.98 ÷ 10 + 321.1 *(approximate)*

889. In what ratio should rice at ₹30/kg be mixed with rice at ₹70/kg to get a mixture worth ₹51/kg?

890. Approximate: 832.02 × 32.98 ÷ 10 + 506.9 *(approximate)*

891. In what ratio should rice at ₹50/kg be mixed with rice at ₹80/kg to get a mixture worth ₹67/kg?

892. Approximate: 434.97 × 34.98 ÷ 10 + 534.1 *(approximate)*

Answers: see the answer key (questions 885–892).

Answers and explanations — Government-exam mixed aptitude

813. **₹5,000** — SI = PRT/100 = 25,000 × 5 × 4 / 100 = 5,000.

814. **144 km; 20 m/s** — D = S × T = 72 × 2 = 144 km. km/h → m/s: × 5/18 → 20 m/s.

815. **A ₹792, B ₹3,564** — Total parts 11; one part = 4,356 ÷ 11 = 396.

816. **₹2,000** — SI = PRT/100 = 5,000 × 8 × 5 / 100 = 2,000.

817. **90 km; 12 1/2 m/s** — D = S × T = 45 × 2 = 90 km. km/h → m/s: × 5/18 → 12 1/2 m/s.

818. **A ₹1,680, B ₹1,008** — Total parts 8; one part = 2,688 ÷ 8 = 336.

819. **₹1,800** — SI = PRT/100 = 15,000 × 6 × 2 / 100 = 1,800.

820. **135 km; 25 m/s** — D = S × T = 90 × 1 1/2 = 135 km. km/h → m/s: × 5/18 → 25 m/s.

821. **A ₹5,334, B ₹3,048** — Total parts 11; one part = 8,382 ÷ 11 = 762.

822. **₹1,600** — SI = PRT/100 = 8,000 × 10 × 2 / 100 = 1,600.

823. **72 km; 10 m/s** — D = S × T = 36 × 2 = 72 km. km/h → m/s: × 5/18 → 10 m/s.

824. **A ₹3,420, B ₹7,695** — Total parts 13; one part = 11,115 ÷ 13 = 855.

825. **₹1,200** — SI = PRT/100 = 12,000 × 5 × 2 / 100 = 1,200.

826. **A ₹2,656, B ₹4,648** — Total parts 11; one part = 7,304 ÷ 11 = 664.

827. **₹19,200** — SI = PRT/100 = 40,000 × 8 × 6 / 100 = 19,200.

828. **A ₹4,672, B ₹2,920** — Total parts 13; one part = 7,592 ÷ 13 = 584.

829. **A ₹1,125, B ₹3,000** — Total parts 11; one part = 4,125 ÷ 11 = 375.

830. **₹24,000** — SI = PRT/100 = 40,000 × 10 × 6 / 100 = 24,000.

831. **67 1/2 km; 25 m/s** — D = S × T = 90 × 3/4 = 67 1/2 km. km/h → m/s: × 5/18 → 25 m/s.

832. **A ₹2,036, B ₹1,527** — Total parts 7; one part = 3,563 ÷ 7 = 509.

833. **₹9,000** — SI = PRT/100 = 25,000 × 6 × 6 / 100 = 9,000.

834. **150 km; 16 2/3 m/s** — D = S × T = 60 × 2 1/2 = 150 km. km/h → m/s: × 5/18 → 16 2/3 m/s.

835. **A ₹1,484, B ₹424** — Total parts 9; one part = 1,908 ÷ 9 = 212.

836. **₹1,000** — SI = PRT/100 = 5,000 × 10 × 2 / 100 = 1,000.

837. **₹35,200** — $A = P(1 + R/100)^n$ = 80,000 × $(120/100)^2$ = ₹1,15,200. CI = A − P = ₹35,200.

838. **4 days** — Total work = LCM(20,5) = 20 units. A does 1/day, B does 4/day; together 5/day. Time = 20/5 = 4 days.

839. **8 8/13 s** — Relative speed = 117 km/h = 32 1/2 m/s. Distance = 200 + 80 = 280 m. Time = 8 8/13 s.

840. **Boat 17 km/h, stream 4 km/h** — Boat = (down + up)/2 = (21 + 13)/2 = 17. Stream = (down − up)/2 = 4.

841. **Son 11, father 22** — Let son = x, father = 2x. (2x + 2) + (x + 2) = 37 → 3x = 33 → x = 11.

842. **Area 2,464 cm², circumference 176 cm** — $A = \pi r^2 = 22/7 \times 28^2 = 2{,}464$; $C = 2\pi r = 176$.

843. **1/13** — 4/52 = 1/13.

844. **56** — Order matters: ${}^nP_r = 8 \times 7 \times \ldots$ (2 factors) = 56.

845. **6 2/3 hours** — Net rate = 1/5 − 1/20 = 3/20 tank/hour, so time = 6 2/3 hours.

846. **11 1/5 s** — Relative speed = 90 km/h = 25 m/s. Distance = 150 + 130 = 280 m. Time = 11 1/5 s.

847. **Boat 10 km/h, stream 6 km/h** — Boat = (down + up)/2 = (16 + 4)/2 = 10. Stream = (down − up)/2 = 6.

848. **Son 10, father 40** — Let son = x, father = 4x. (4x + 10) + (x + 10) = 70 → 5x = 50 → x = 10.

849. **2,772 cm³** — $V = \frac{1}{3}\pi r^2 h = 2{,}772$.

850. **1/4** — $C(n, k)/2^n = 1/4 = 1/4$.

851. **21** — Order does not matter: ${}^nC_r = 7!/(2! \cdot 5!) = 21$.

852. **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 724 mod 4 = 4 → 1.

853. **12 hours** — Net rate = 1/8 − 1/24 = 1/12 tank/hour, so time = 12 hours.

854. **6 2/15 s** — Relative speed = 135 km/h = 37 1/2 m/s. Distance = 100 + 130 = 230 m. Time = 6 2/15 s.

855. **Boat 13 km/h, stream 2 km/h** — Boat = (down + up)/2 = (15 + 11)/2 = 13. Stream = (down − up)/2 = 2.

856. **Son 19, father 38** — Let son = x, father = 2x. (2x + 5) + (x + 5) = 67 → 3x = 57 → x = 19.

857. **1,386 cm³** — $V = \pi r^2 h = 22/7 \times 49 \times 9 = 1{,}386$.

858. **1/2** — 26/52 = 1/2.

859. **495** — Order does not matter: ${}^{n}C_{r}$ = 12!/(4!·8!) = 495.

860. **2** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 799 mod 4 = 3 → 2.

861. **11 5/13 s** — Relative speed = 117 km/h = 32 1/2 m/s. Distance = 240 + 130 = 370 m. Time = 11 5/13 s.

862. **Boat 17 km/h, stream 1 km/h** — Boat = (down + up)/2 = (18 + 16)/2 = 17. Stream = (down − up)/2 = 1.

863. **Son 20, father 80** — Let son = x, father = 4x. (4x + 4) + (x + 4) = 108 → 5x = 100 → x = 20.

864. **Area 406 m², perimeter 86 m** — A = l × w; P = 2(l + w).

865. **11,880** — Order matters: ${}^{n}P_{r}$ = 12 × 11 × … (4 factors) = 11,880.

866. **4** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 22 mod 4 = 2 → 4.

867. **2** — 97,832 = 9 × 10,870 + 2. Shortcut: digit sum mod 9.

868. **3** — Division and multiplication left to right first: 6 ÷ 3 = 2; × 2 = 4. Then 12 + 4 − 13 = 3.

869. **1 : 4** — Alligation cross: (60 − 56) : (56 − 40) = 4 : 16 = 1 : 4.

870. **≈ 3068 (exact 3070.18)** — Round each number: 888 × 33 ÷ 10 + 138 ≈ 3068.4.

871. **11 : 4** — Alligation cross: (80 − 36) : (36 − 20) = 44 : 16 = 11 : 4.

872. **≈ 2383 (exact 2384.99)** — Round each number: 711 × 24 ÷ 10 + 677 ≈ 2383.4.

873. **2 : 13** — Alligation cross: (60 − 56) : (56 − 30) = 4 : 26 = 2 : 13.

874. **≈ 1206 (exact 1205.30)** — Round each number: 337 × 24 ÷ 10 + 397 ≈ 1205.8.

875. **11 : 4** — Alligation cross: (70 − 48) : (48 − 40) = 22 : 8 = 11 : 4.

876. **≈ 2030 (exact 2029.33)** — Round each number: 356 × 48 ÷ 10 + 321 ≈ 2029.8.

877. **≈ 2775 (exact 2774.08)** — Round each number: 413 × 47 ÷ 10 + 834 ≈ 2775.1.

878. **11 : 9** — Alligation cross: (70 − 59) : (59 − 50) = 11 : 9 = 11 : 9.

879. **≈ 1206 (exact 1206.12)** — Round each number: 286 × 26 ÷ 10 + 462 ≈ 1205.6.

880. **23 : 27** — Alligation cross: (90 − 67) : (67 − 40) = 23 : 27 = 23 : 27.

881. **≈ 1623 (exact 1623.75)** — Round each number: 396 × 30 ÷ 10 + 435 ≈ 1623.0.

882. **23 : 37** — Alligation cross: (90 − 67) : (67 − 30) = 23 : 37 = 23 : 37.

883. **≈ 2653 (exact 2654.77)** — Round each number: 579 × 36 ÷ 10 + 569 ≈ 2653.4.

884. **33 : 7** — Alligation cross: (70 − 37) : (37 − 30) = 33 : 7 = 33 : 7.

885. **53 : 7** — Alligation cross: (80 − 27) : (27 − 20) = 53 : 7 = 53 : 7.

886. **≈ 2381 (exact 2379.47)** — Round each number: 892 × 22 ÷ 10 + 419 ≈ 2381.4.

887. **19 : 11** — Alligation cross: (70 − 51) : (51 − 40) = 19 : 11 = 19 : 11.

888. **≈ 1982 (exact 1980.59)** — Round each number: 692 × 24 ÷ 10 + 321 ≈ 1981.8.

889. **19 : 21** — Alligation cross: (70 − 51) : (51 − 30) = 19 : 21 = 19 : 21.

890. **≈ 3253 (exact 3250.90)** — Round each number: 832 × 33 ÷ 10 + 507 ≈ 3252.6.

891. **13 : 17** — Alligation cross: (80 − 67) : (67 − 50) = 13 : 17 = 13 : 17.

892. **≈ 2056 (exact 2055.63)** — Round each number: 435 × 35 ÷ 10 + 534 ≈ 2056.5.

Advanced mental-math challenge — 10 sets

✎ Set 61: Advanced mental-math challenge · Beginner · suggested time 6 min

893. Find the last digit of 27^{897}.

894. Find the last digit of 3^{420}.

895. Find the last digit of 17^{112}.

896. 102^2

897. Find the last digit of 9^{649}.

898. Find the last digit of 8^{243}.

899. 968^2

900. Find the last digit of 4^{347}.

Answers: see the answer key (questions 893–900).

✎ Set 62: Advanced mental-math challenge · Beginner · suggested time 6 min

901. Find the last digit of 17^{584}.

902. 101^2

903. Find the last digit of 8^{436}.

904. 986^2

905. Find the last digit of 17^{79}.

906. 988^2

907. Find the last digit of 4^{574}.

908. Find the last digit of 27^{616}.

Answers: see the answer key (questions 901–908).

✎ Set 63: Advanced mental-math challenge - Beginner - suggested time 6 min

909. Find the last digit of 9^{183}.

910. 983^2

911. Find the last digit of 22^{599}.

912. $1{,}038^2$

913. Find the last digit of 13^{152}.

914. Find the last digit of 17^{329}.

915. 99^2

916. Find the last digit of 8^{764}.

Answers: see the answer key (questions 909–916).

✎ Set 64: Advanced mental-math challenge - Intermediate - suggested time 10 min

917. 111^2

918. Find the last digit of 2^{593}.

919. Find the last digit of 38^{658}.

920. 94^2

921. Find the last digit of 33^{809}.

922. Find the last digit of 9^{166}.

923. 86^2

924. Find the last digit of 38^{467}.

Answers: see the answer key (questions 917–924).

✎ Set 65: Advanced mental-math challenge · Intermediate · suggested time 10 min

925. Find the last digit of 8^{308}.

926. $1{,}005^2$

927. Find the last digit of 9^{542}.

928. 112^2

929. Find the last digit of 8^{33}.

930. Find the last digit of 27^{921}.

931. 984^2

932. Find the last digit of 7^{445}.

Answers: see the answer key (questions 925–932).

✎ Set 66: Advanced mental-math challenge · Intermediate · suggested time 10 min

933. Find the last digit of 8^{298}.

934. 91^2

935. Find the last digit of 2^{261}.

936. 92^2

937. Find the last digit of 13^{998}.

938. 110^2

939. Find the last digit of 4^{402}.

940. Find the last digit of 9^{180}.

Answers: see the answer key (questions 933–940).

✎ Set 67: Advanced mental-math challenge · Intermediate · suggested time 10 min

941. Find the last digit of 9^{322}.

942. 113^2

943. Find the last digit of 8^{771}.

944. Find the last digit of 3^{580}.

945. Find the last digit of 27^{124}.

946. 97^2

947. Find the last digit of 27^{456}.

948. 87^2

Answers: see the answer key (questions 941–948).

✎ Set 68: Advanced mental-math challenge · Advanced · suggested time 16 min

949. Solve: 39x + 16y = 183, 16x + 39y = 367

950. Estimate √140 to 2 decimal places (approximate). *(approximate)*

951. Solve $x^2 + 5x - 6 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

952. 704 × 416

953. 647^2 (duplex method)

954. 644 ÷ 14

955. 4,171 ÷ 97

956. 47 × 41 (working base 50)

Answers: see the answer key (questions 949–956).

✎ Set 69: Advanced mental-math challenge · Advanced · suggested time 16 min

957. Estimate √161 to 2 decimal places (approximate). *(approximate)*

958. Solve $x^2 - 10x + 24 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

959. 708 × 445

960. 156^2 (duplex method)

961. 66,198 ÷ 1012

962. 325 ÷ 8

963. Find the recurring block of 1/19.

964. 408 × 395 (working base 400)

Answers: see the answer key (questions 957–964).

✎ Set 70: Advanced mental-math challenge · Advanced · suggested time 16 min

965. Estimate √135 to 2 decimal places (approximate). *(approximate)*

966. Solve $x^2 + 3x + 2 = 0$ using $2x + b = \pm\sqrt{(b^2 - 4c)}$

967. 610 × 781

968. 798^2 (duplex method)

969. 8,766 ÷ 121

970. 16,606 ÷ 998

971. 395 × 407 (working base 400)

972. What day of the week was/is 12 September 1974?

Answers: see the answer key (questions 965–972).

Answers and explanations — Advanced mental-math challenge

893. **7** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 897 mod 4 = 1 → 7.

894. **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 420 mod 4 = 4 → 1.

895. **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 112 mod 4 = 4 → 1.

896. **10,404** — Base 100, deviation +2 Left: 102 + 2 = 104 Right: $2^2 = 4$ (needs 2 digit(s)) Answer = 10,404

897. **9** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 649 mod 2 = 1 → 9.

898. **2** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 243 mod 4 = 3 → 2.

899. **9,37,024** — Base 1,000, deviation −32 Left: 968 − 32 = 936 Right: $32^2 = 1024$ (needs 3 digit(s), carry 1) Answer = 9,37,024

900. **4** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 347 mod 2 = 1 → 4.

901. **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 584 mod 4 = 4 → 1.

902. **10,201** — Base 100, deviation +1 Left: 101 + 1 = 102 Right: $1^2 = 1$ (needs 2 digit(s)) Answer = 10,201

903. **6** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 436 mod 4 = 4 → 6.

904. **9,72,196** — Base 1,000, deviation −14 Left: 986 − 14 = 972 Right: $14^2 = 196$ (needs 3 digit(s)) Answer = 9,72,196

905. **3** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 79 mod 4 = 3 → 3.

906. **9,76,144** — Base 1,000, deviation −12 Left: 988 − 12 = 976 Right: $12^2 = 144$ (needs 3 digit(s)) Answer = 9,76,144

907. **6** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 574 mod 2 = 2 → 6.

908. **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 616 mod 4 = 4 → 1.

909. **9** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 183 mod 2 = 1 → 9.

910. **9,66,289** — Base 1,000, deviation −17 Left: 983 − 17 = 966 Right: $17^2 = 289$ (needs 3 digit(s)) Answer = 9,66,289

911. **8** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 599 mod 4 = 3 → 8.

912. **10,77,444** — Base 1,000, deviation +38 Left: 1038 + 38 = 1076 Right: $38^2 = 1444$ (needs 3 digit(s), carry 1) Answer = 10,77,444

913. **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 152 mod 4 = 4 → 1.

914. **7** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 329 mod 4 = 1 → 7.

915. **9,801** — Base 100, deviation −1 Left: 99 − 1 = 98 Right: $1^2 = 1$ (needs 2 digit(s)) Answer = 9,801

916. **6** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 764 mod 4 = 4 → 6.

917. **12,321** — Base 100, deviation +11 Left: 111 + 11 = 122 Right: $11^2 = 121$ (needs 2 digit(s), carry 1) Answer = 12,321

918. **2** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 593 mod 4 = 1 → 2.

919. **4** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 658 mod 4 = 2 → 4.

920. **8,836** — Base 100, deviation −6 Left: 94 − 6 = 88 Right: $6^2 = 36$ (needs 2 digit(s)) Answer = 8,836

921. **3** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 809 mod 4 = 1 → 3.

922. **1** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 166 mod 2 = 2 → 1.

923. **7,396** — Base 100, deviation −14 Left: 86 − 14 = 72 Right: $14^2 = 196$ (needs 2 digit(s), carry 1) Answer = 7,396

924. **2** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 467 mod 4 = 3 → 2.

925. **6** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 308 mod 4 = 4 → 6.

926. **10,10,025** — Base 1,000, deviation +5 Left: 1005 + 5 = 1010 Right: $5^2 = 25$ (needs 3 digit(s)) Answer = 10,10,025

927. **1** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 542 mod 2 = 2 → 1.

928. **12,544** — Base 100, deviation +12 Left: 112 + 12 = 124 Right: $12^2 = 144$ (needs 2 digit(s), carry 1) Answer = 12,544

929. **8** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 33 mod 4 = 1 → 8.

930. **7** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 921 mod 4 = 1 → 7.

931. **9,68,256** — Base 1,000, deviation −16 Left: 984 − 16 = 968 Right: $16^2 = 256$ (needs 3 digit(s)) Answer = 9,68,256

932. **7** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 445 mod 4 = 1 → 7.

933. **4** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 298 mod 4 = 2 → 4.

934. **8,281** — Base 100, deviation −9 Left: 91 − 9 = 82 Right: $9^2 = 81$ (needs 2 digit(s)) Answer = 8,281

935. **2** — Last digits of powers of 2 repeat: 2, 4, 8, 6 (cycle 4). 261 mod 4 = 1 → 2.

936. **8,464** — Base 100, deviation −8 Left: 92 − 8 = 84 Right: $8^2 = 64$ (needs 2 digit(s)) Answer = 8,464

937. **9** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 998 mod 4 = 2 → 9.

938. **12,100** — Base 100, deviation +10 Left: 110 + 10 = 120 Right: 10^2 = 100 (needs 2 digit(s), carry 1) Answer = 12,100

939. **6** — Last digits of powers of 4 repeat: 4, 6 (cycle 2). 402 mod 2 = 2 → 6.

940. **1** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 180 mod 2 = 2 → 1.

941. **1** — Last digits of powers of 9 repeat: 9, 1 (cycle 2). 322 mod 2 = 2 → 1.

942. **12,769** — Base 100, deviation +13 Left: 113 + 13 = 126 Right: 13^2 = 169 (needs 2 digit(s), carry 1) Answer = 12,769

943. **2** — Last digits of powers of 8 repeat: 8, 4, 2, 6 (cycle 4). 771 mod 4 = 3 → 2.

944. **1** — Last digits of powers of 3 repeat: 3, 9, 7, 1 (cycle 4). 580 mod 4 = 4 → 1.

945. **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 124 mod 4 = 4 → 1.

946. **9,409** — Base 100, deviation −3 Left: 97 − 3 = 94 Right: 3^2 = 9 (needs 2 digit(s)) Answer = 9,409

947. **1** — Last digits of powers of 7 repeat: 7, 9, 3, 1 (cycle 4). 456 mod 4 = 4 → 1.

948. **7,569** — Base 100, deviation −13 Left: 87 − 13 = 74 Right: 13^2 = 169 (needs 2 digit(s), carry 1) Answer = 7,569

949. **x = 1, y = 9** — Add: 55(x + y) = 550 → x + y = 10. Subtract: 23(x − y) = −184 → x − y = −8. Solve: x = 1, y = 9.

950. **≈ 11.83 (true value 11.832)** — Nearest perfect square 144 = 12^2. $\sqrt{n} \approx 12 + (140 - 144) \div (2 \times 12)$ = 11.833.

951. **x = 1 or −6** — Differential: 2x + 5. Discriminant = 25 − -24 = 49, √ = 7. 2x + 5 = ±7.

952. **2,92,864** — Column 1 (from right): 4×6 = 24 → write 4, carry 2 Column 2 (from right): 4×1 + 0×6 = 4, + carry 2 = 6 → write 6, carry 0 Column 3 (from right): 4×4 + 0×1 + 7×6 = 58 → write 8, carry 5 Column 4 (from right): 0×4 + 7×1 = 7, + carry 5 = 12 → write 2, carry 1 Column 5 (from right): 7×4 = 28, + carry 1 = 29 → write 9, carry 2 Answer = 2,92,864

953. **4,18,609** — Duplexes from left: 36, 48, 100, 56, 49 Combine with carries (right to left) → 4,18,609

954. **Q = 46, R = 0** — Divisor 14 = 10 + 4; transposed flags: −4 Column values after applying flags: [6, -20] | [84] Resolve negatives/carries → quotient 46, remainder 0

955. **Q = 43, R = 0** — Divisor 97 = 100 − 3; write complement digits 03 Split dividend: 41 | 71 Column totals after multiplying by the complement: [4, 1] | [19, 4] Resolve carries and remainder → quotient 43, remainder 0

956. **1,927** — Working base 50 = 5 × 10. Deviations −3, −9. Left: (47 −9) × 5 = 190. Right: −3 × −9 = 27. 190 × 10 + 27 = 1,927.

957. **≈ 12.69 (true value 12.689)** — Nearest perfect square 169 = 13^2. $\sqrt{n} \approx 13 + (161 - 169) \div (2 \times 13)$ = 12.692.

958. **x = 6 or 4** — Differential: 2x − 10. Discriminant = 100 − 96 = 4, √ = 2. 2x − 10 = ±2.

959. **3,15,060** — Column 1 (from right): 8×5 = 40 → write 0, carry 4 Column 2 (from right): 8×4 + 0×5 = 32, + carry 4 = 36 → write 6, carry 3 Column 3 (from right): 8×4 + 0×4 + 7×5 = 67, + carry 3 = 70 → write 0, carry 7 Column 4 (from right): 0×4 + 7×4 = 28, + carry 7 = 35 → write 5, carry 3 Column 5 (from right): 7×4 = 28, + carry 3 = 31 → write 1, carry 3 Answer = 3,15,060

960. **24,336** — Duplexes from left: 1, 10, 37, 60, 36 Combine with carries (right to left) → 24,336

961. **Q = 65, R = 418** — Divisor 1012 = 1,000 + 12; transposed flags: −0, −1, −2 Column values after applying flags: [6, 6] | [-5, -9, -4] Resolve negatives/carries → quotient 65, remainder 418

962. **Q = 40, R = 5** — Divisor 8 = 10 − 2; write complement digits 2 Split dividend: 32 | 5 Column totals after multiplying by the complement: [3, 8] | [21] Resolve carries and remainder → quotient 40, remainder 5

963. **0.(052631578947368421)** — Ekadhika (one more than 1) = 2 Start with 1 on the right; repeatedly multiply by 2, write the unit digit, carry the rest Recurring block (18 digits): 052631578947368421

964. **1,61,160** — Working base 400 = 4 × 100. Deviations +8, −5. Left: (408 −5) × 4 = 1612. Right: 8 × −5 = −40. 1612 × 100 + −40 = 1,61,160.

965. **≈ 11.62 (true value 11.619)** — Nearest perfect square 144 = 12^2. $\sqrt{n} \approx 12 + (135 - 144) \div (2 \times 12)$ = 11.625.

966. **x = −1 or −2** — Differential: 2x + 3. Discriminant = 9 − 8 = 1, √ = 1. 2x + 3 = ±1.

967. **4,76,410** — Column 1 (from right): 0×1 = 0 → write 0, carry 0 Column 2 (from right): 0×8 + 1×1 = 1 → write 1, carry 0 Column 3 (from right): 0×7 + 1×8 + 6×1 = 14 → write 4, carry 1 Column 4 (from right): 1×7 + 6×8 = 55, + carry 1 = 56 → write 6, carry 5 Column 5 (from right): 6×7 = 42, + carry 5 = 47 → write 7, carry 4 Answer = 4,76,410

968. **6,36,804** — Duplexes from left: 49, 126, 193, 144, 64 Combine with carries (right to left) → 6,36,804

969. **Q = 72, R = 54** — Divisor 121 = 100 + 21; transposed flags: −2, −1 Column values after applying flags: [8, -9] | [16, 15] Resolve negatives/carries → quotient 72, remainder 54

970. **Q = 16, R = 638** — Divisor 998 = 1,000 − 2; write complement digits 002 Split dividend: 16 | 606 Column totals after multiplying by the complement: [1, 6] | [6, 2, 18] Resolve carries and remainder → quotient 16, remainder 638

971. **1,60,765** — Working base 400 = 4 × 100. Deviations −5, +7. Left: (395 +7) × 4 = 1608. Right: −5 × 7 = −35. 1608 × 100 + −35 = 1,60,765.

972. **Thursday** — Use odd days: count year codes, month codes and leap years (see Part IX), then take the total mod 7.